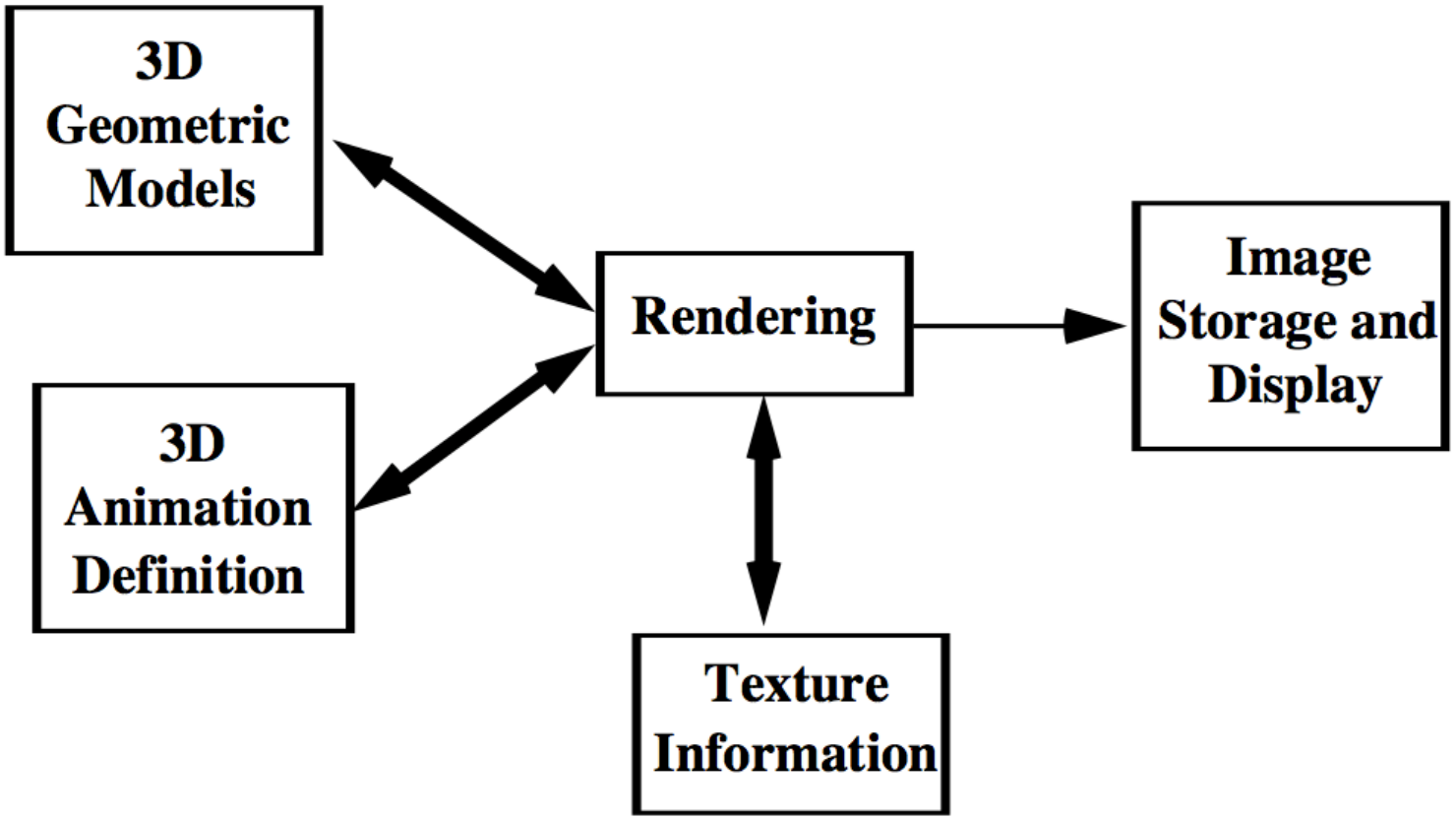
The Graphics Process



The Graphics Process



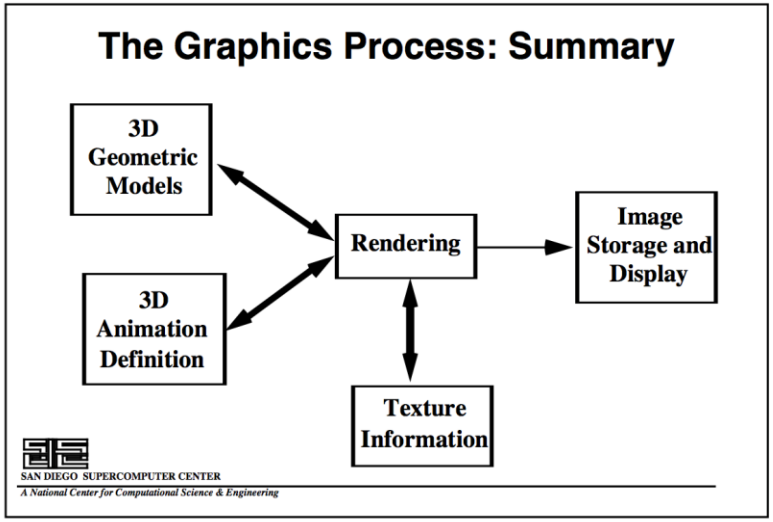
The Graphics Process: Summary



The Graphics Process: Summary

3D Geometric Models

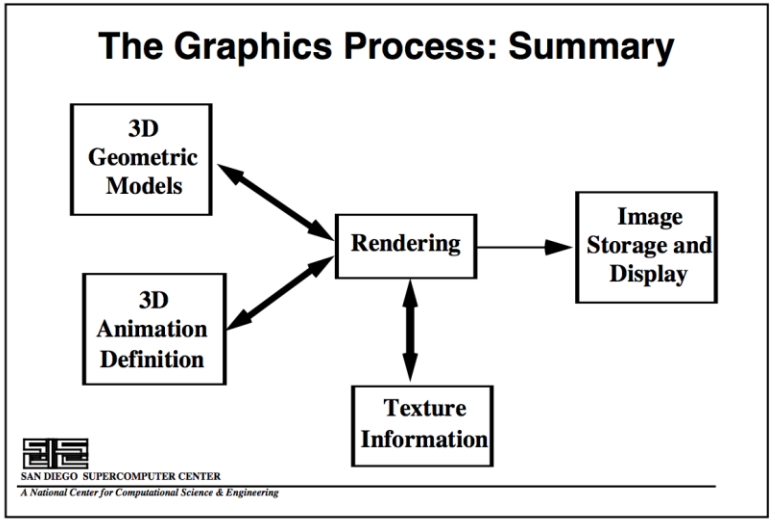
- Represent the shape and structure of objects (e.g., a chair, car, or character).
- These models define the geometry (vertices, edges, surfaces) used in a scene.



The Graphics Process: Summary

3D Animation Definition

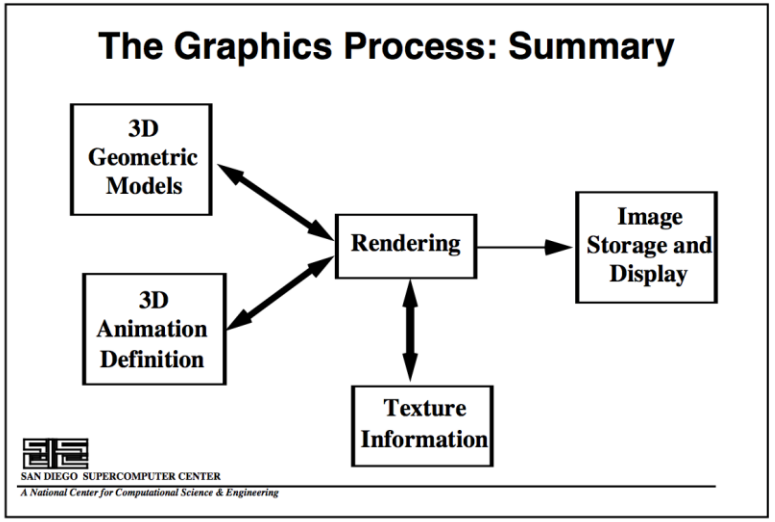
- Defines motion and transformations of the geometric models over time. (e.g., walking cycle of a character, rotation of a wheel, or camera movement).



The Graphics Process: Summary

Texture Information

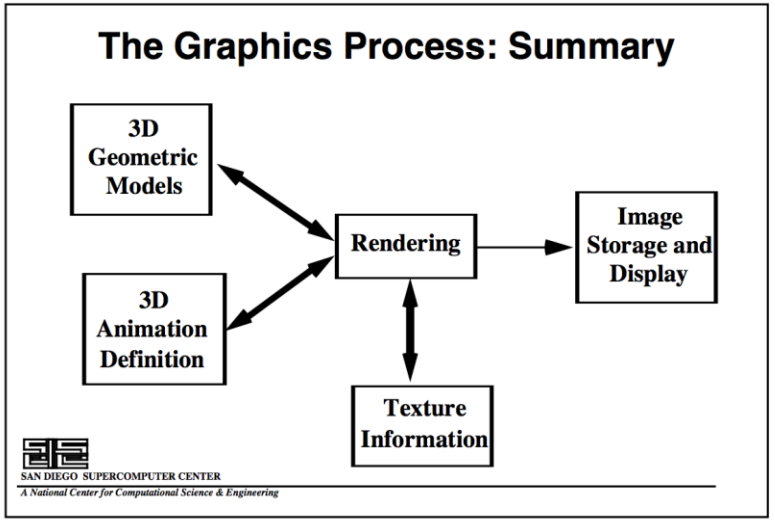
- Adds surface details (color, patterns, materials) to the models. (e.g., a brick wall texture on a flat surface makes it look like real bricks).
- This is connected directly to Rendering to enrich visual realism.



The Graphics Process: Summary

Rendering

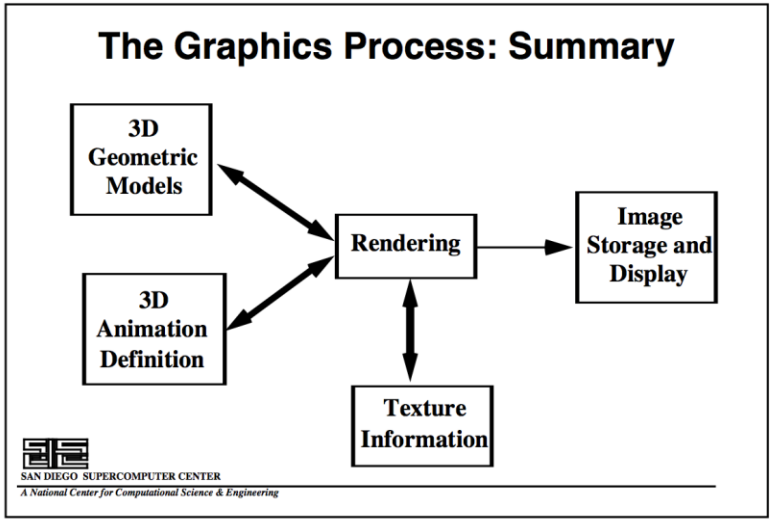
- The central process that takes models, animations, and textures and converts them into 2D images or video frames.
- Involves lighting, shading, camera perspective, reflections, and shadows.
- Rendering produces the final visual output.



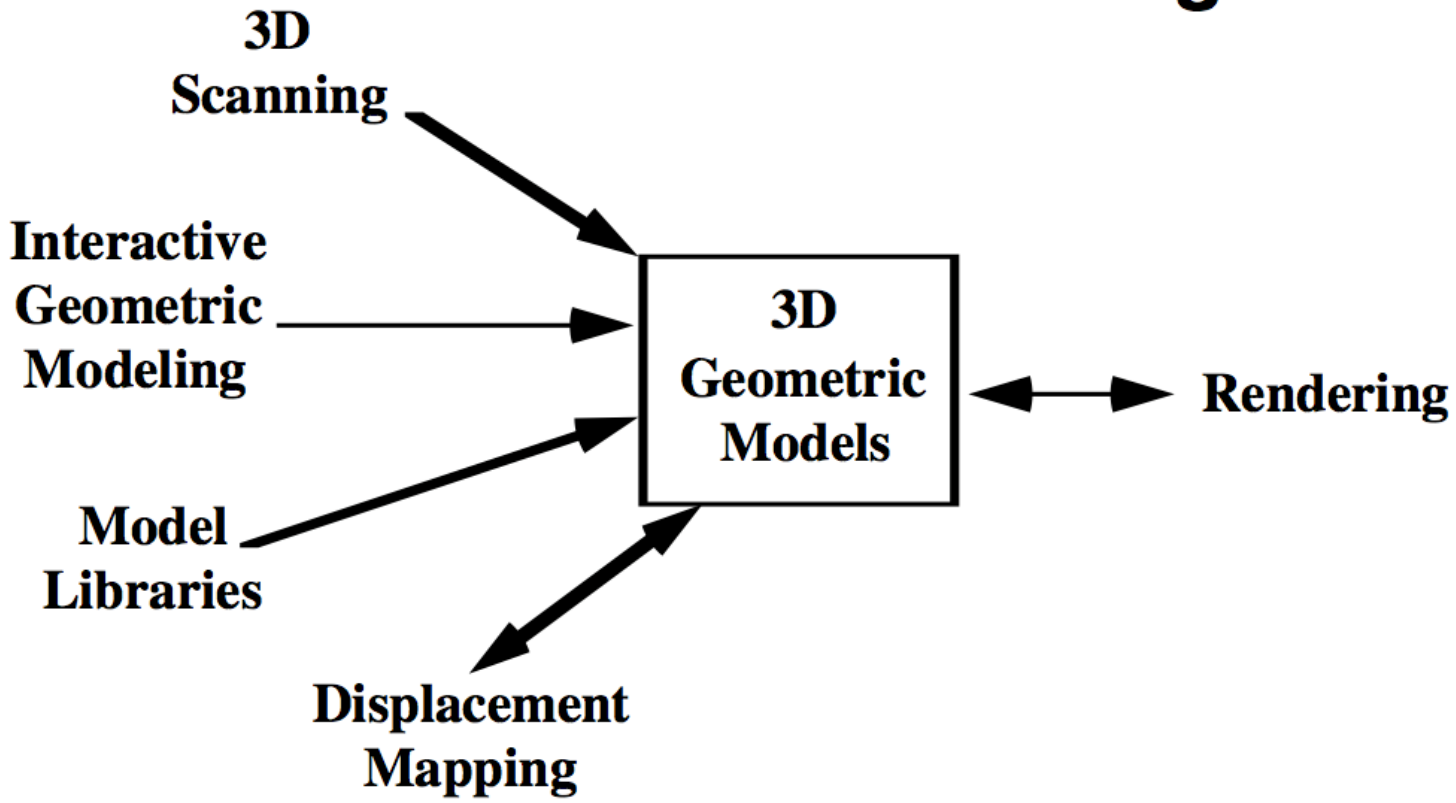
The Graphics Process: Summary

Image Storage and Display

- After rendering, the resulting images/frames are:
  - Stored (in memory, disk, or video files).
  - Displayed (on screens, VR devices, projectors).
- This is what users finally see.



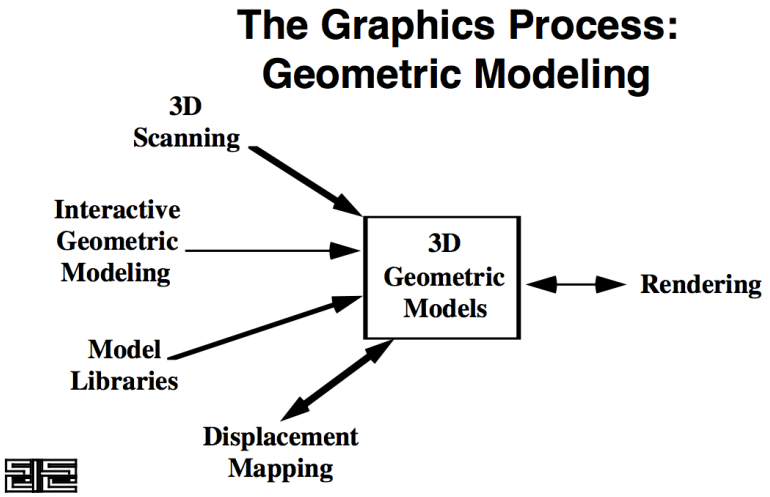
The Graphics Process: Geometric Modeling



The Graphics Process: Geometric Modeling

Inputs to 3D Geometric Models

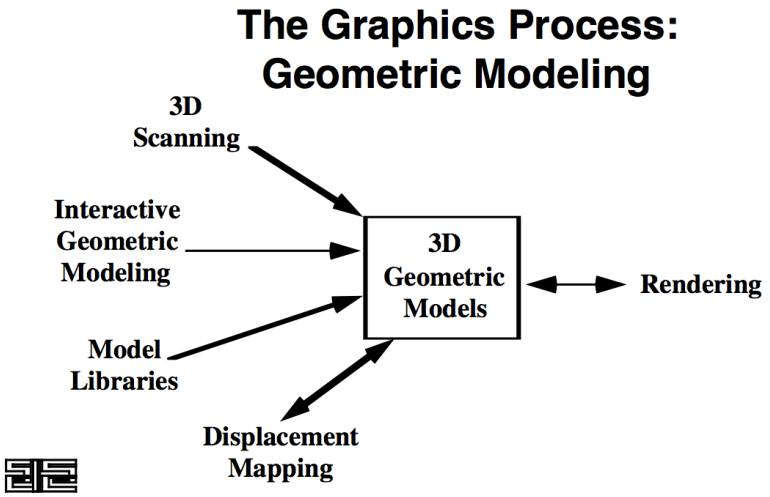
- **3D Scanning**: Uses scanners to capture the shape and structure of real-world objects and convert them into digital 3D models.



The Graphics Process: Geometric Modeling

Inputs to 3D Geometric Models

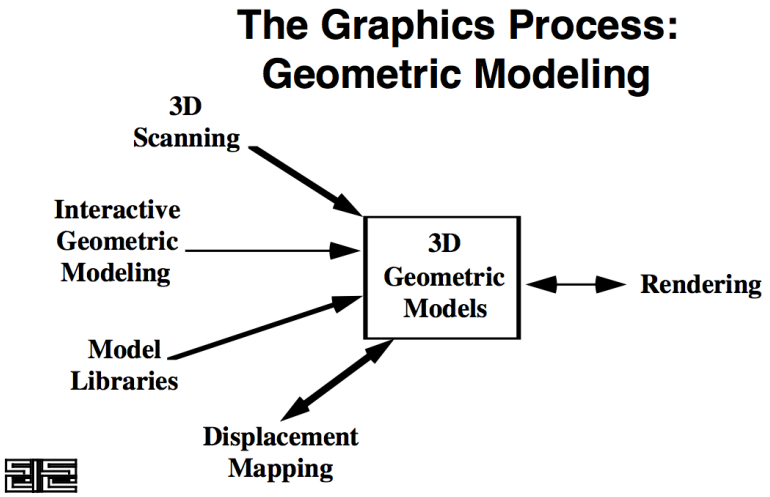
- **Interactive Geometric Modeling**: Artists or designers create models directly using modeling software (e.g., Blender, Maya, CAD tools).



The Graphics Process: Geometric Modeling

Inputs to 3D Geometric Models

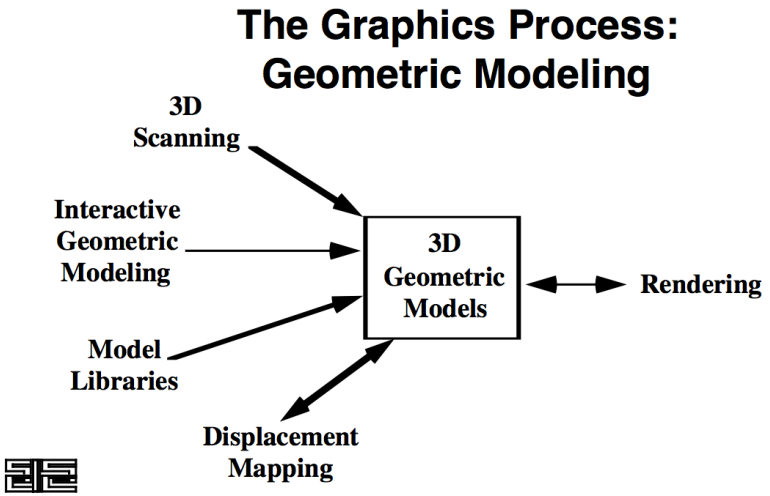
- **Model Libraries**: Pre-existing 3D models stored in databases can be imported and reused.



The Graphics Process: Geometric Modeling

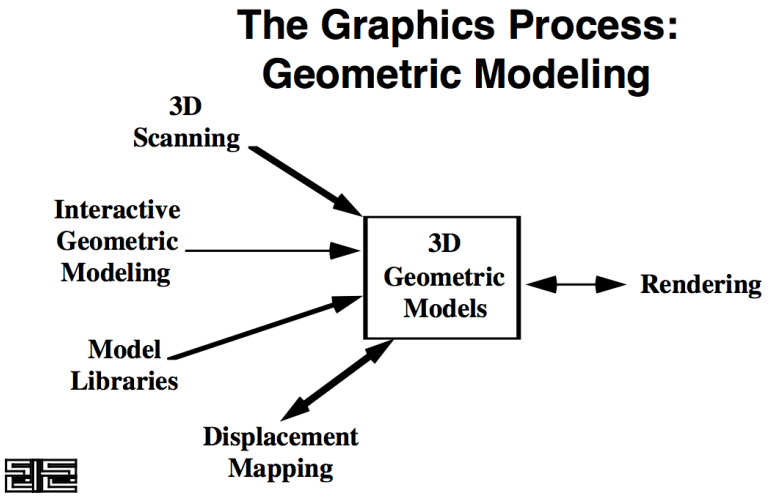
Inputs to 3D Geometric Models

- **Displacement Mapping**: Adds fine surface detail to models by displacing geometry based on texture maps (useful for realistic surfaces like rocks, skin, etc.).



The Graphics Process: Geometric Modeling

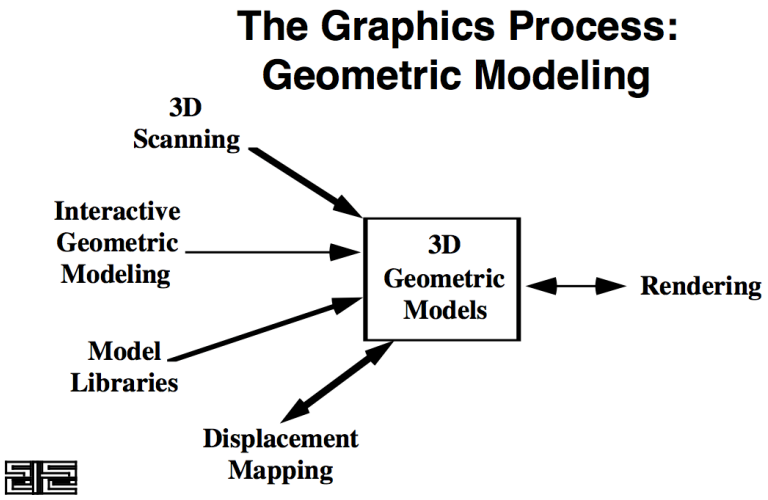
- **3D Geometric Models**
  - This is the central block where all input methods contribute. These models represent the **shape and structure** of objects in a scene.



The Graphics Process: Geometric Modeling

Rendering

- The created 3D models are sent to the **rendering process**, where lighting, shading, textures, and camera perspectives are applied to generate the final 2D image or animation.

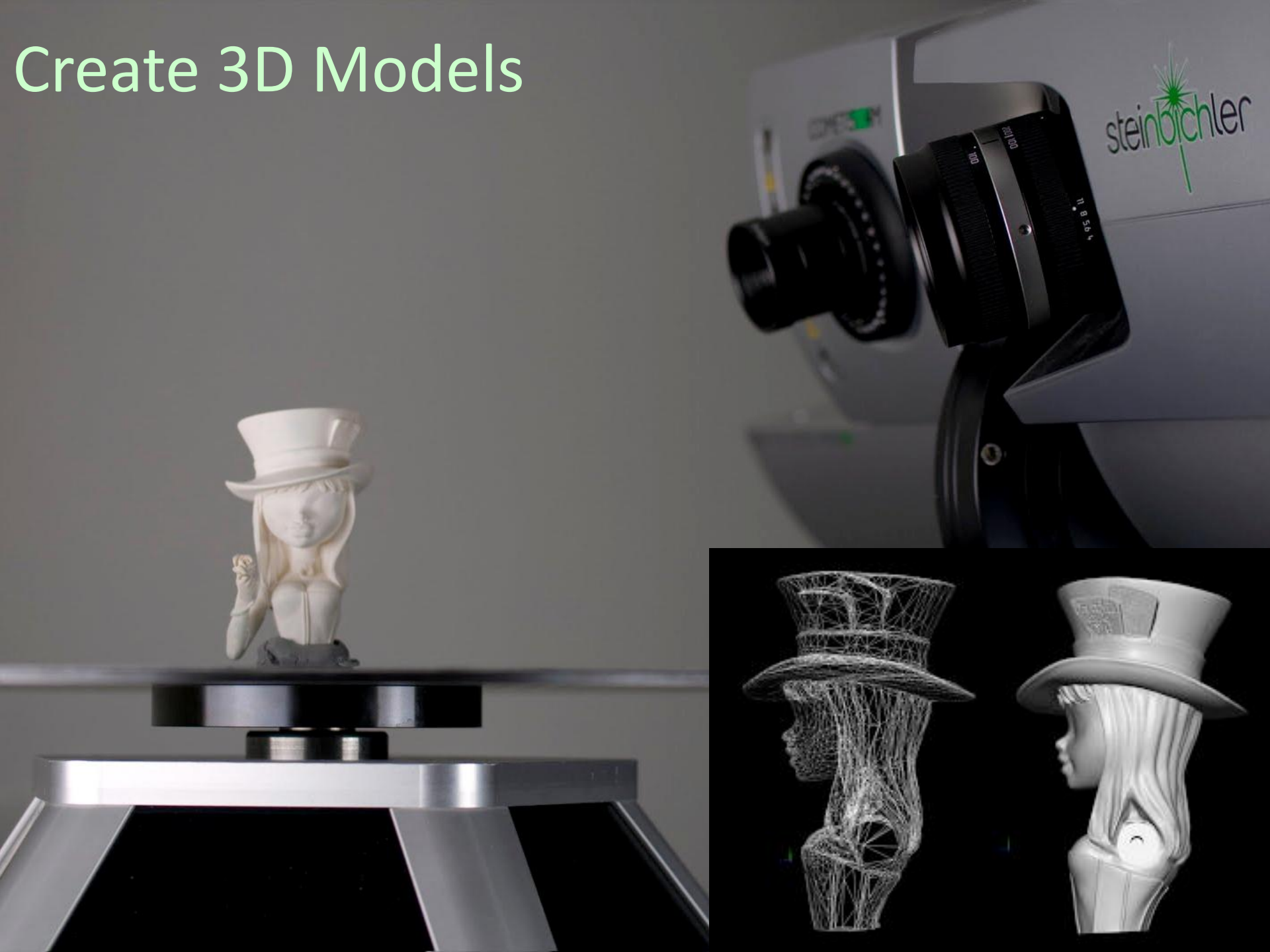


The **two-way arrow**: rendering may provide feedback to improve or adjust the geometric models (e.g., refining details when the render doesn't look correct).

Create 3D Models



Create 3D Models



Create 3D Models







Create 3D Models

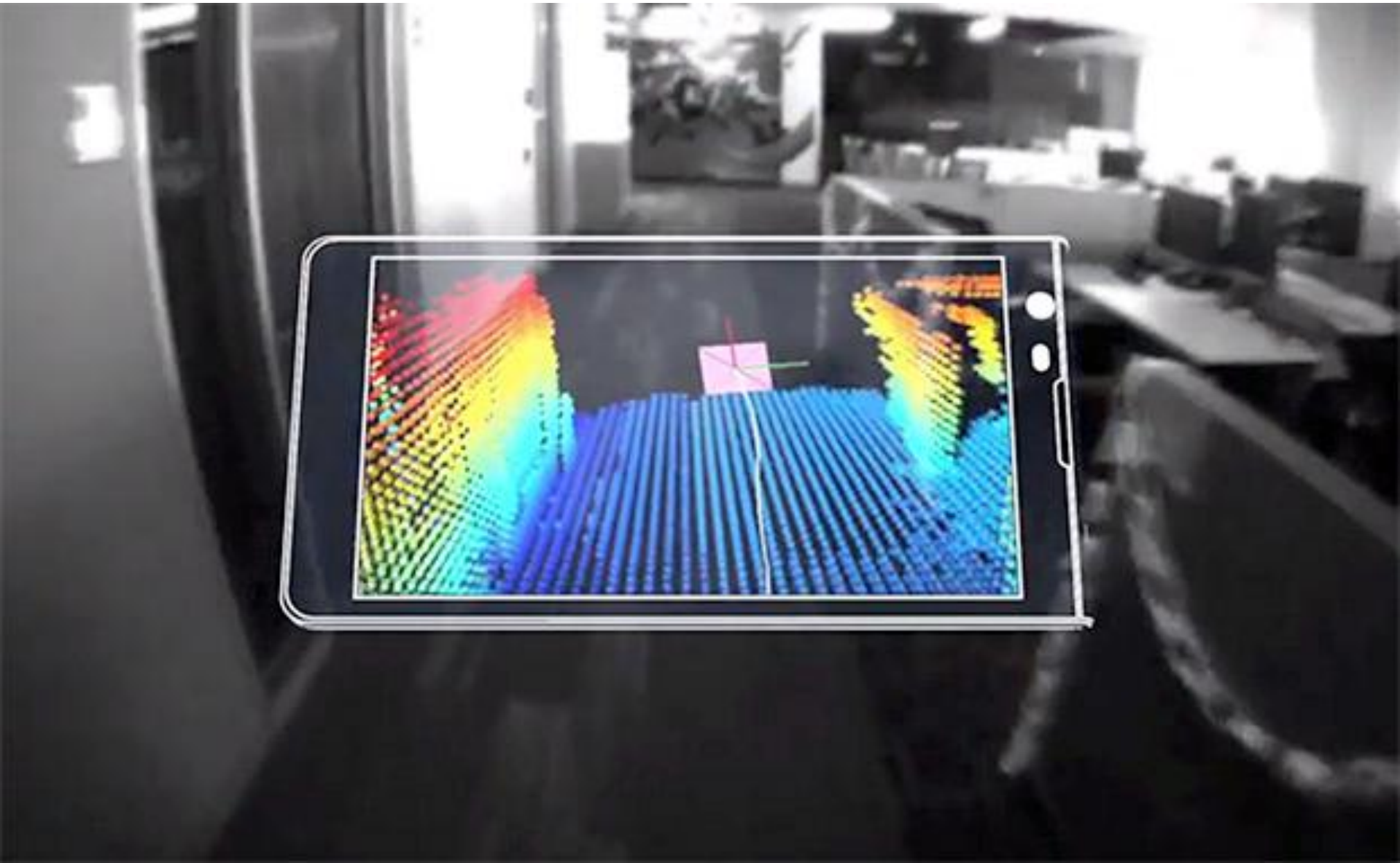
AUTODESK®
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Shoot a loop of about 20 sequential photographs in small increments about your subject.

Shoot another loop from a different angle if you can.

Create 3D Models



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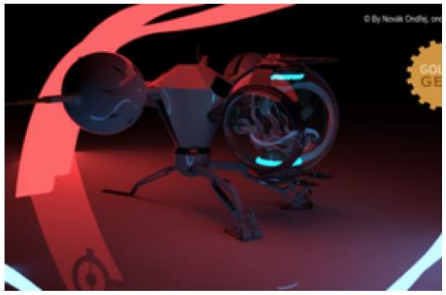
Models per page: **20** | 40 | 100



AUDI R8
453 69783



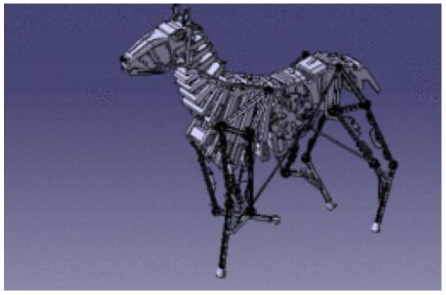
Apple iPhone 4
149 34134



Oblivion copter
74 33638



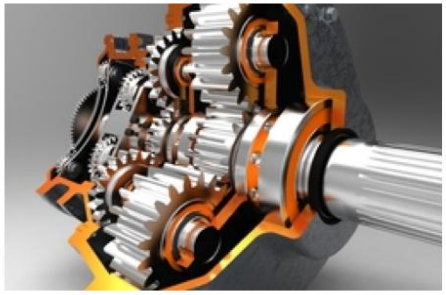
Colt 1911-A1 Model Goverment Pistol
163 33250



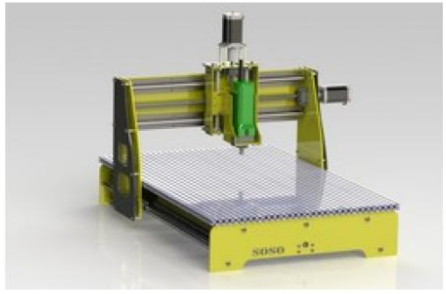
mechanical horse



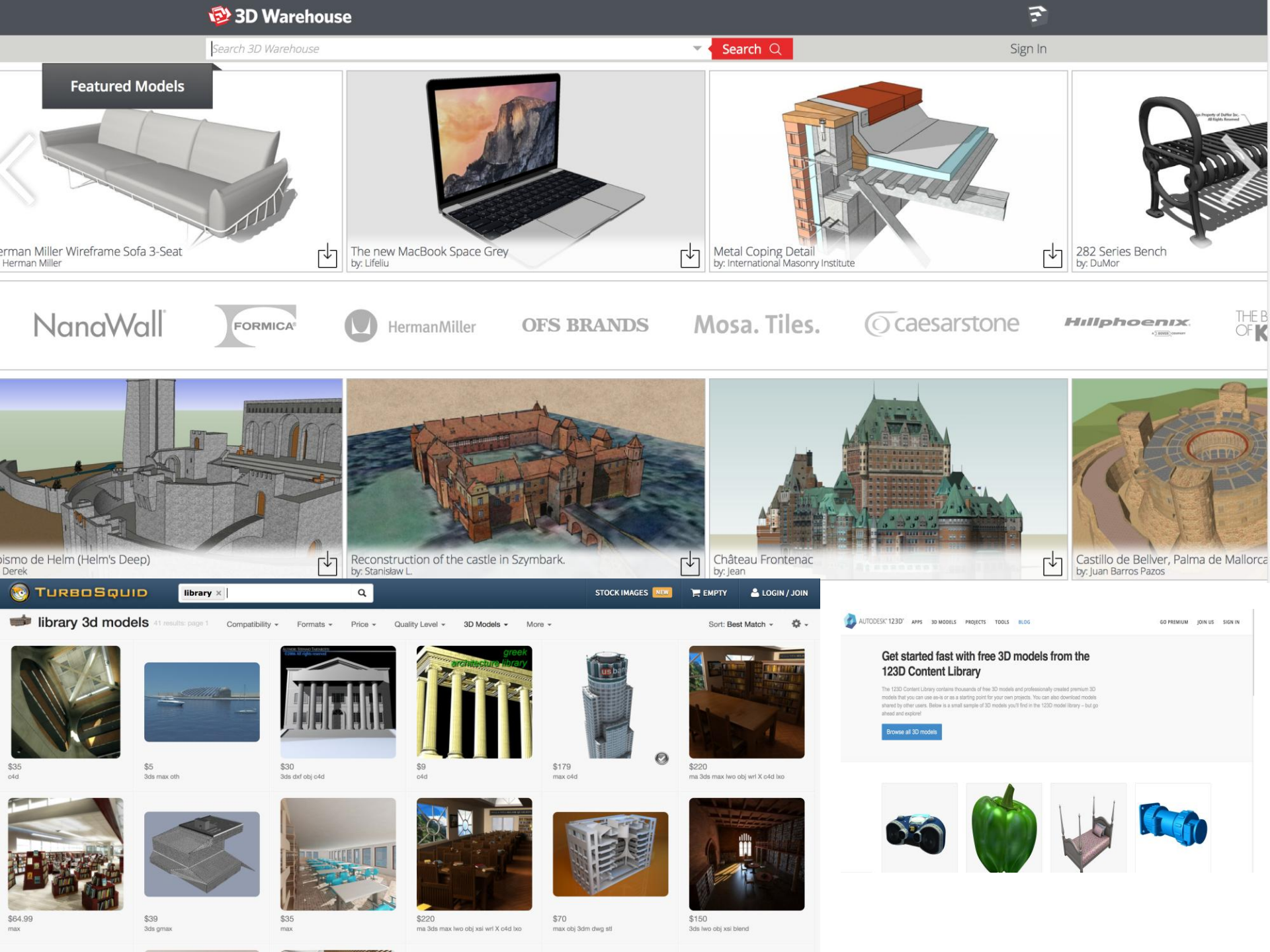
MQ-1 Predator UAV



Planetary Gearbox



CNC 3 axis


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caesarstone
Hillphoenix
ismo de Helm (Helm's Deep)
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Reconstruction of the castle in Szymbark.
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by: Jean
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Formats
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Quality Level
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Sort: Best Match
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Get started fast with free 3D models from the 123D Content Library
The 123D Content Library contains thousands of free 3D models and professionally created premium 3D models that you can use as-is or as a starting point for your own projects. You can also download models shared by other users. Below is a small sample of 3D models you'll find in the 123D model library – but go ahead and explore!
Browse all 3D models

List of 3D modeling software

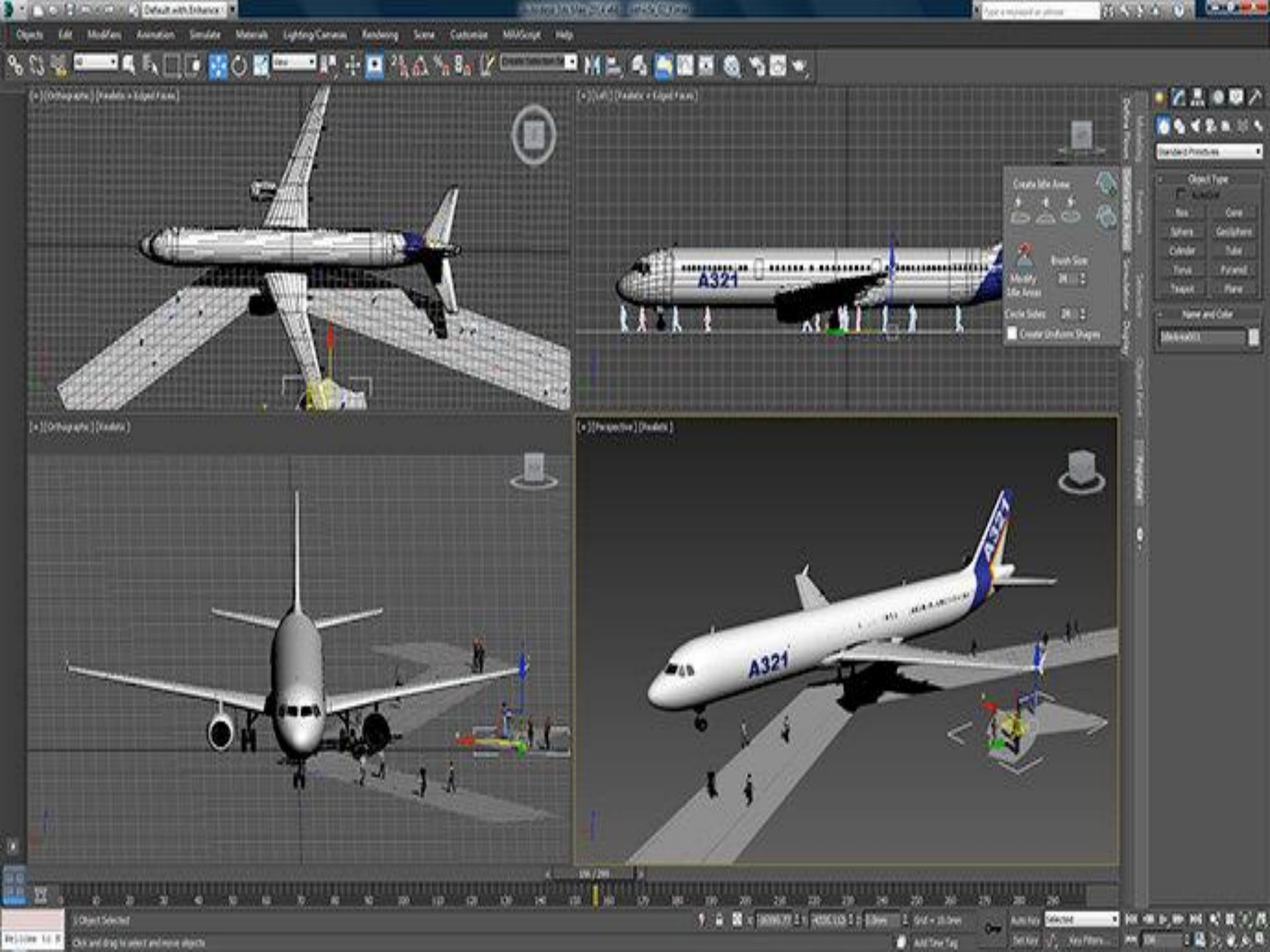
Title	License	3D rendering support
Autodesk 123D	Freeware	No
Autodesk 3ds Max	Commercial software	Yes
Autodesk Inventor	Commercial software	Yes
Autodesk Maya	Commercial software	Yes

64 more rows



List of 3D modeling software - Wikipedia, the free ...

https://en.wikipedia.org/wiki/List_of_**3D_modeling_software** Wikipedia

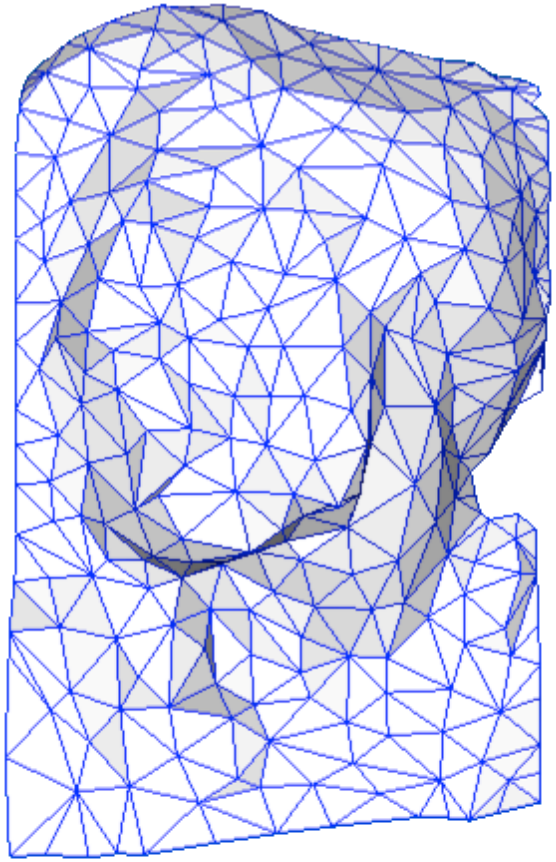

A321
A321
A321


Blender* [D:\BillyOposto\@Sources\Models\player\render\billy-op
Zélia Duncan - Mãos Atadas (4:08)
File Add Render Help
Default
Scene
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billy-sapato.L
billy-sapato.R
cabelo
calca
camisa
dentes.baixo
dentes.cima
olho.L
Armature
Armature
Transform
Location:
X: 0.022
Y: -0.794
Z: -12.477
Rotation:
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Y: 0°
Z: 0°
Scale:
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Y: 6.676
Z: 6.676
Rotation Mode:
XYZ Euler
Delta Transform
Transform Locks
Relations
Layers:
Parent:
Object
Pass Index: 0
Groups
Add to Group
Display
Type:
Textured
Bounds
Sphere
Blender 2.5
2.57.0
r36147
www.img-nation.de
Interaction:
Preset
Links
Donations
Release Log
Manual
Blender Website
User Community
Python API Reference
Recent
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View Frame Playback
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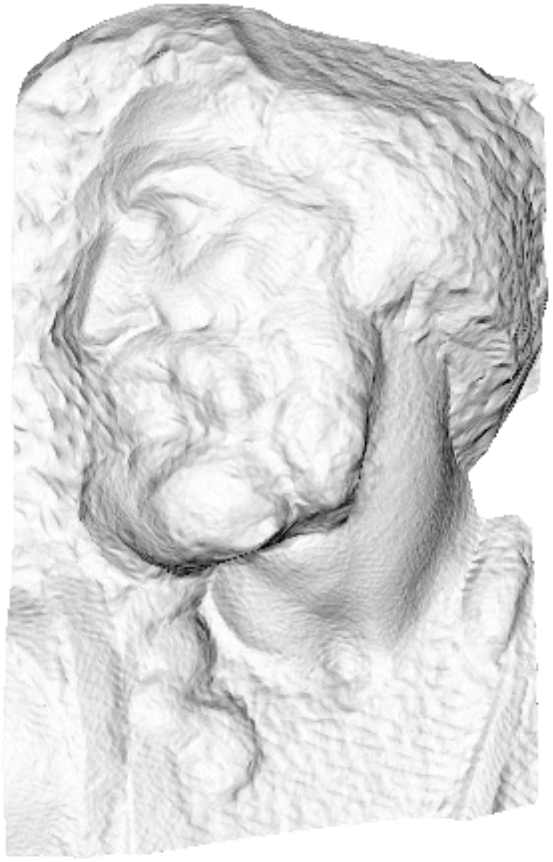
Normal Mapping



original mesh
4M triangles



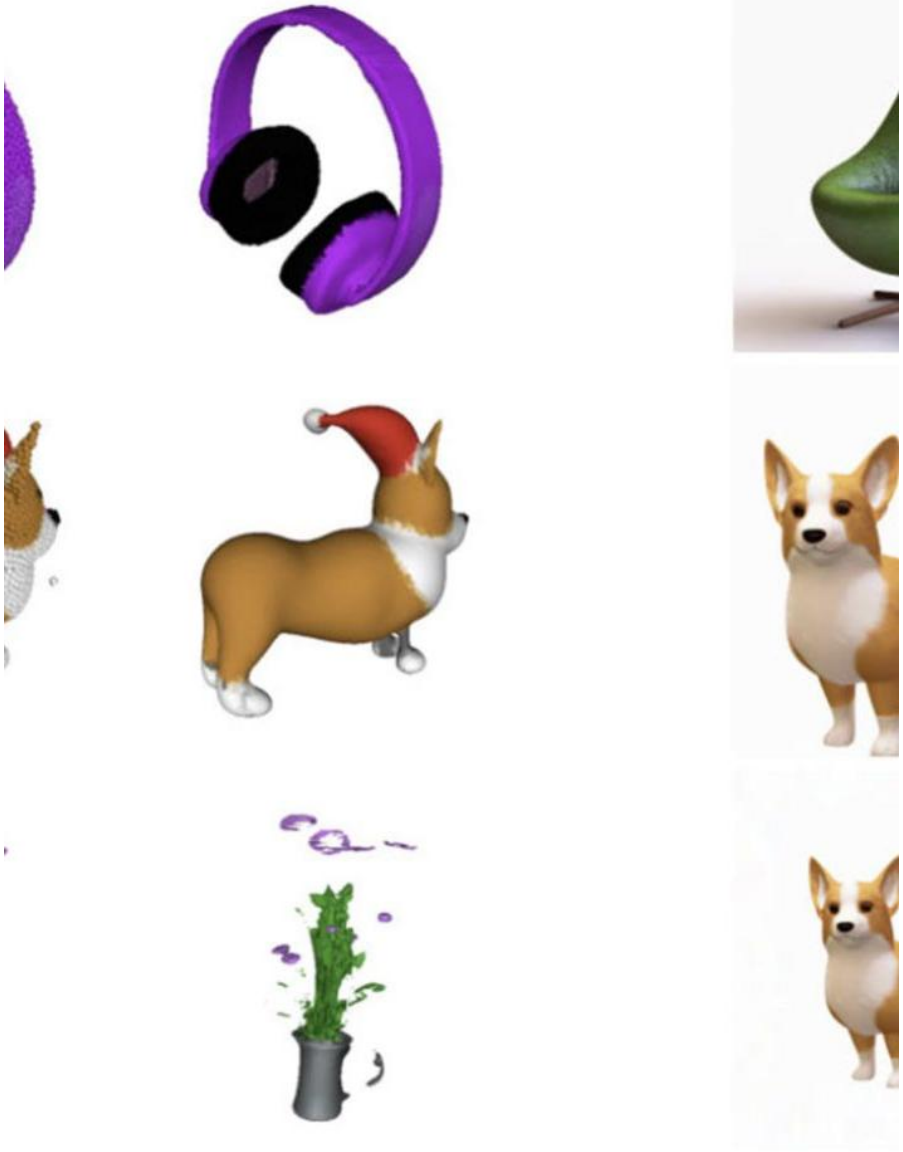
simplified mesh
500 triangles



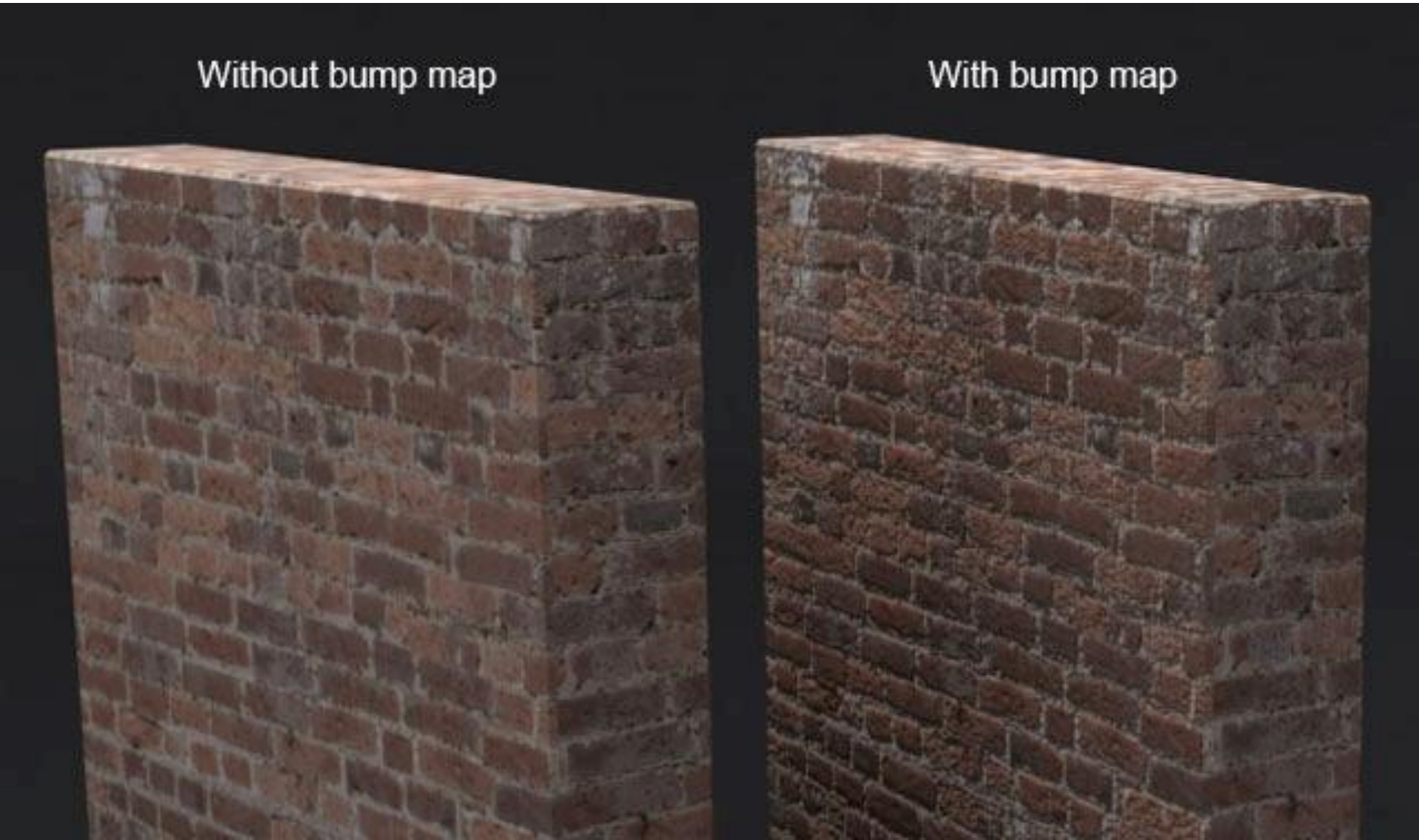
simplified mesh
and normal mapping
500 triangles



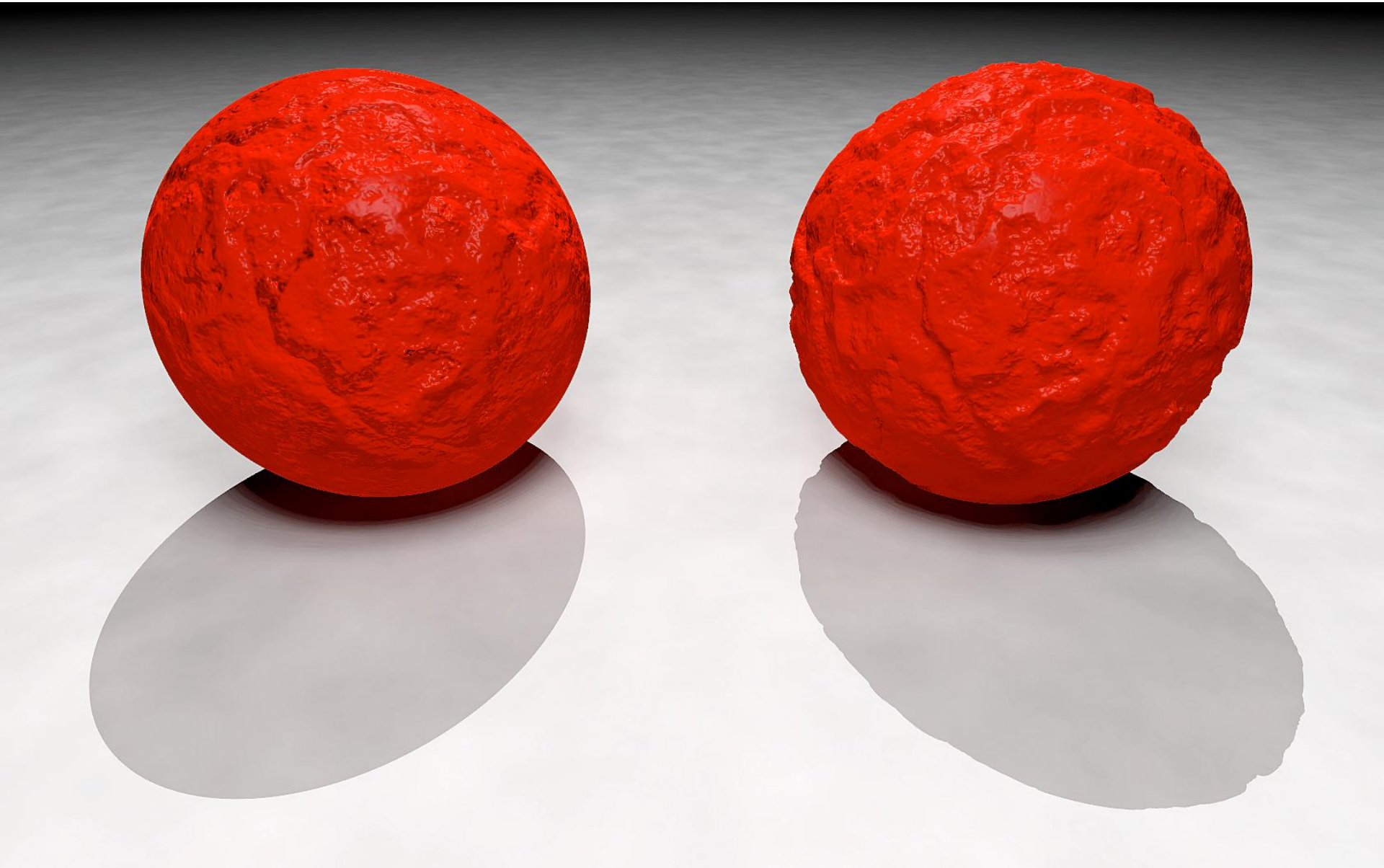

AI
3D
MODEL



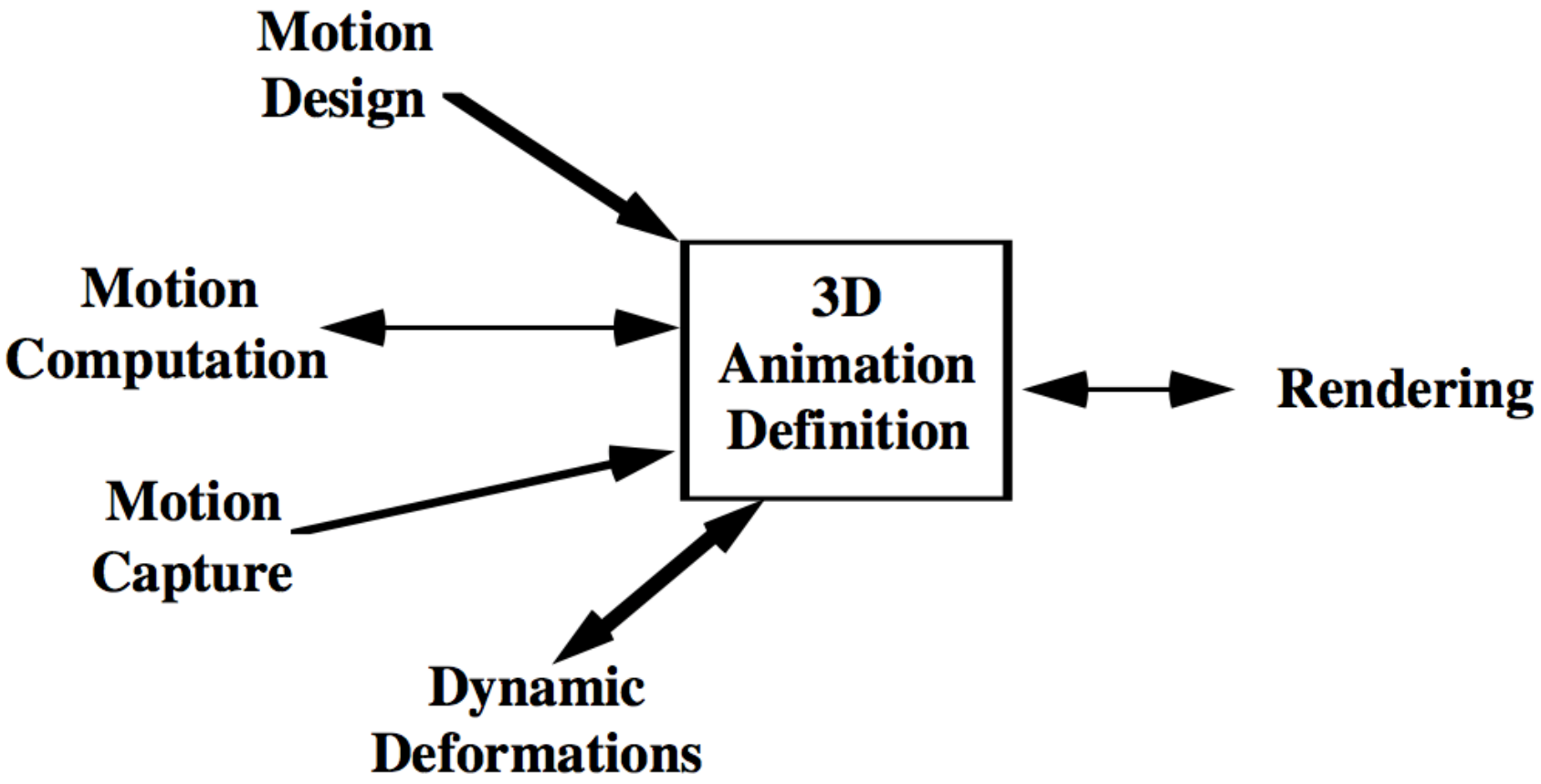
Bump Mapping



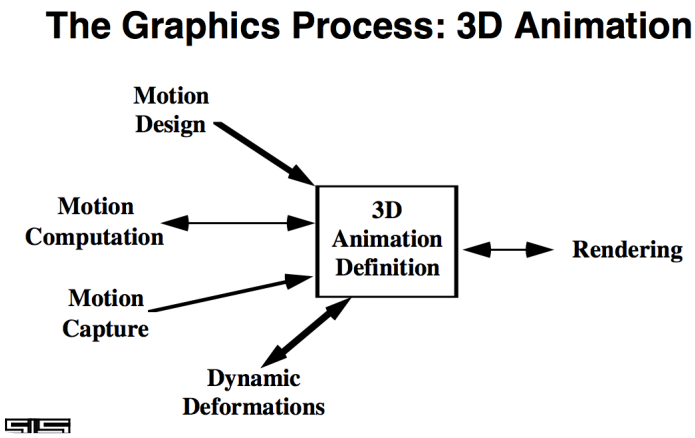
Bump Mapping



The Graphics Process: 3D Animation



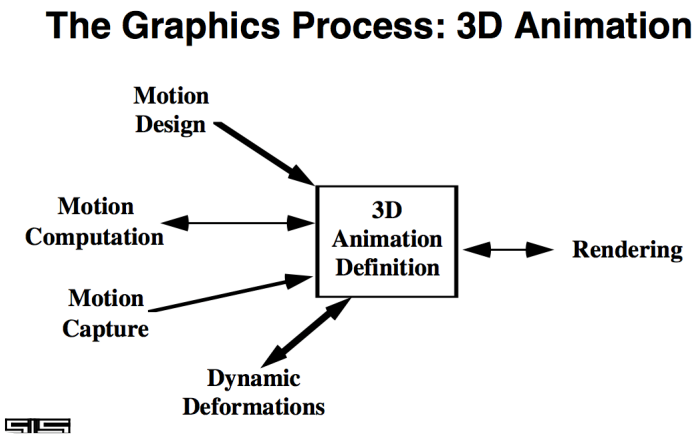
The Graphics Process: 3D Animation



3D Animation Definition

- This is the core process where the motion and behavior of 3D models are specified.
- It tells how objects move, deform, or interact over time.

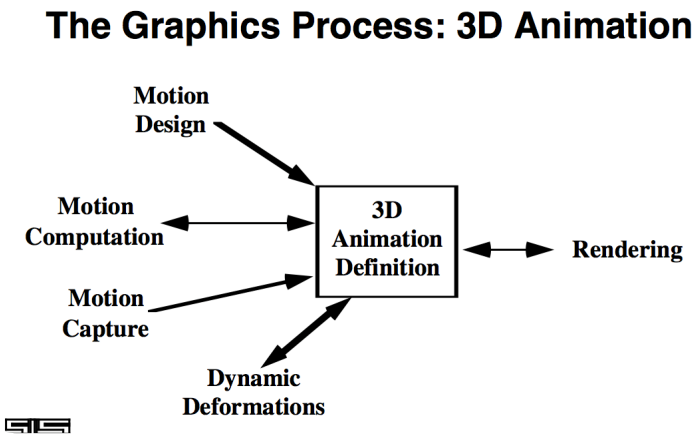
The Graphics Process: 3D Animation



Inputs to 3D Animation Definition

- Motion Design
  - Manual creation of animations by animators.
  - Example: designing a character's walk cycle or an explosion effect in software like Blender or Maya.

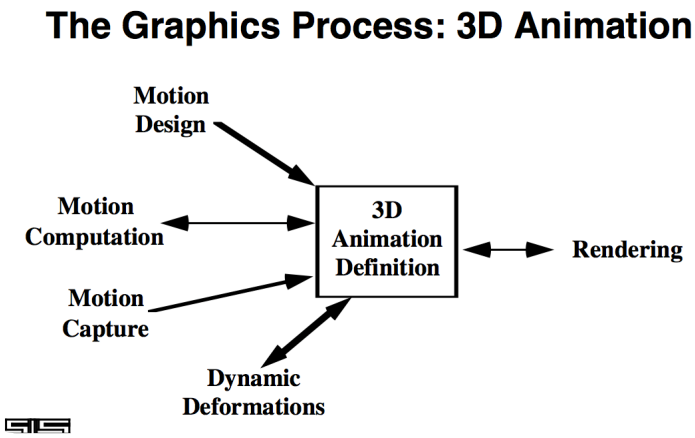
The Graphics Process: 3D Animation



Motion Computation

- Uses algorithms, physics simulations, or AI to calculate movements.
  - Example: simulating how a ball bounces or how cloth flows in the wind.

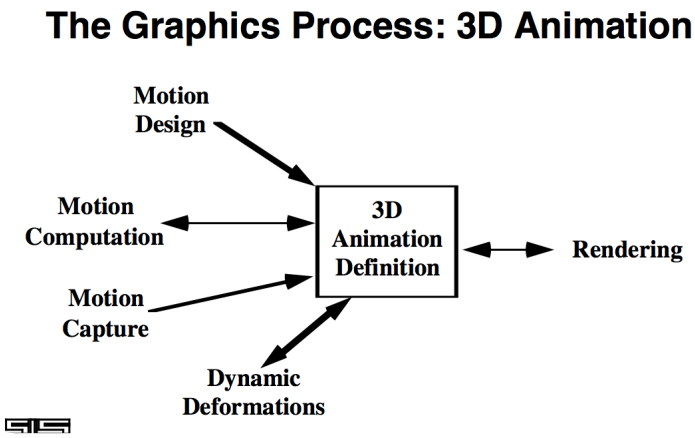
The Graphics Process: 3D Animation



Motion Capture

- Records real human or object movement using sensors or cameras and maps it to 3D models.
  - Example: actors wearing motion capture suits to animate characters in movies (like characters in The Avatar Movie).

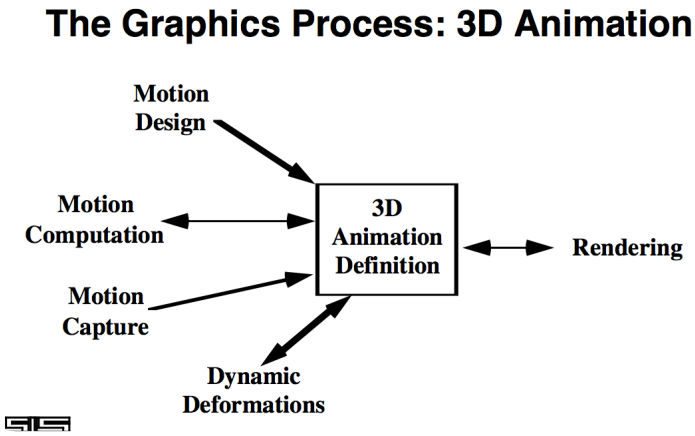
The Graphics Process: 3D Animation



Dynamic Deformations

- Handles changes in the shape of objects during animation.
  - Example: muscles bulging when a character flexes, or a car body denting in a crash simulation.

The Graphics Process: 3D Animation

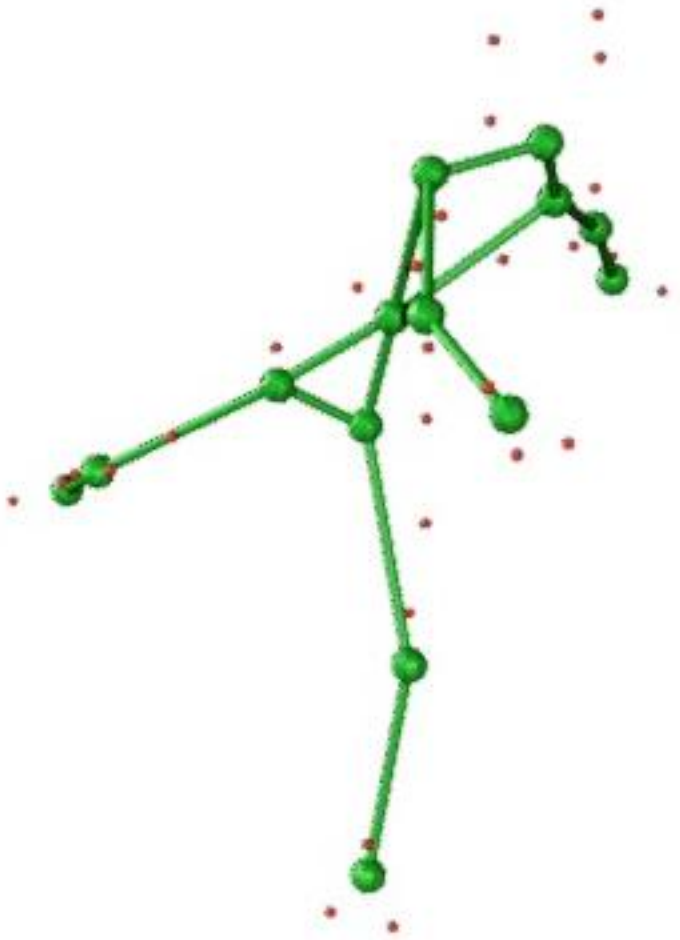


Rendering

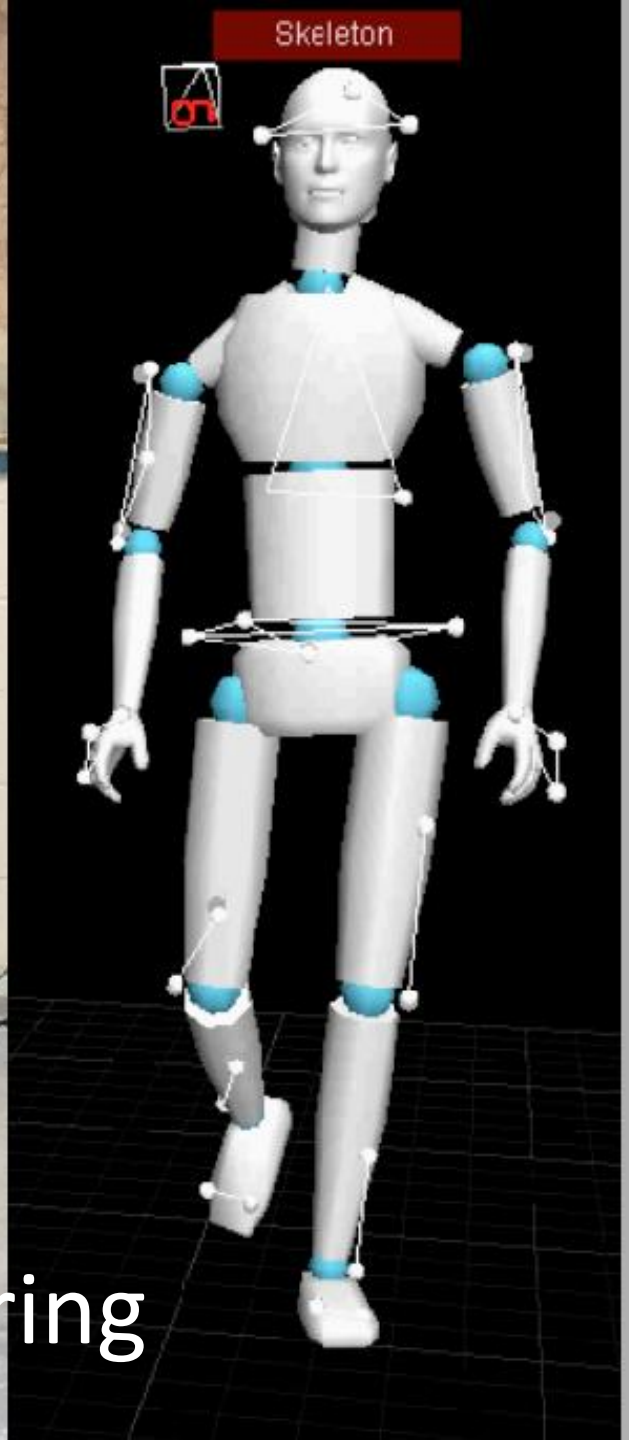
- Once the 3D animation is defined, it goes into rendering, where lighting, shading, and texturing are applied to produce the final frames.
- The two-way arrow means rendering can give feedback- if motion doesn't look realistic, animators or algorithms can refine it.



Motion capturing









Motion capturing


Motion capturing


Motion capturing

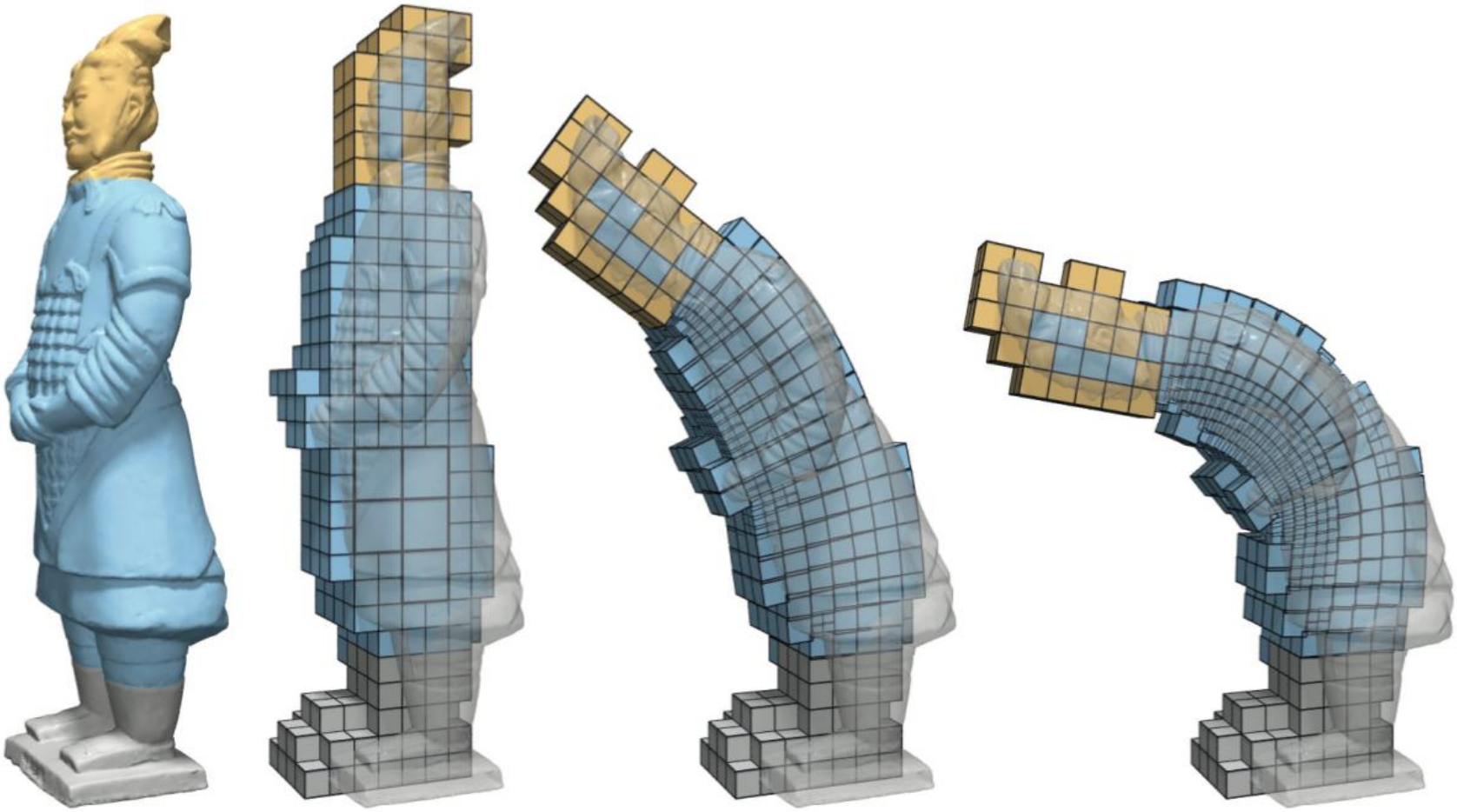
Human Motion Computation

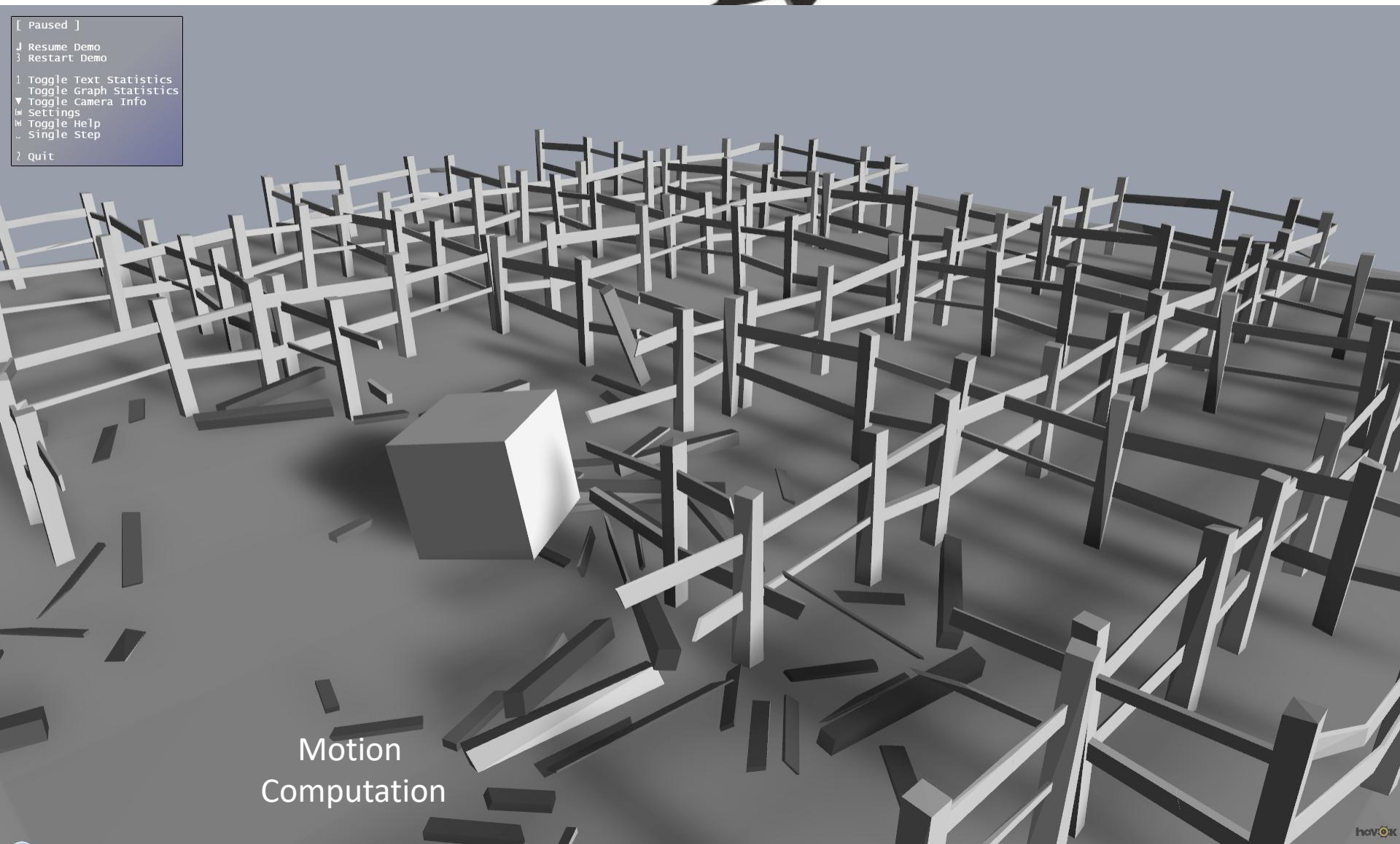


Human 3D Motion Computation from a varying Number of Cameras

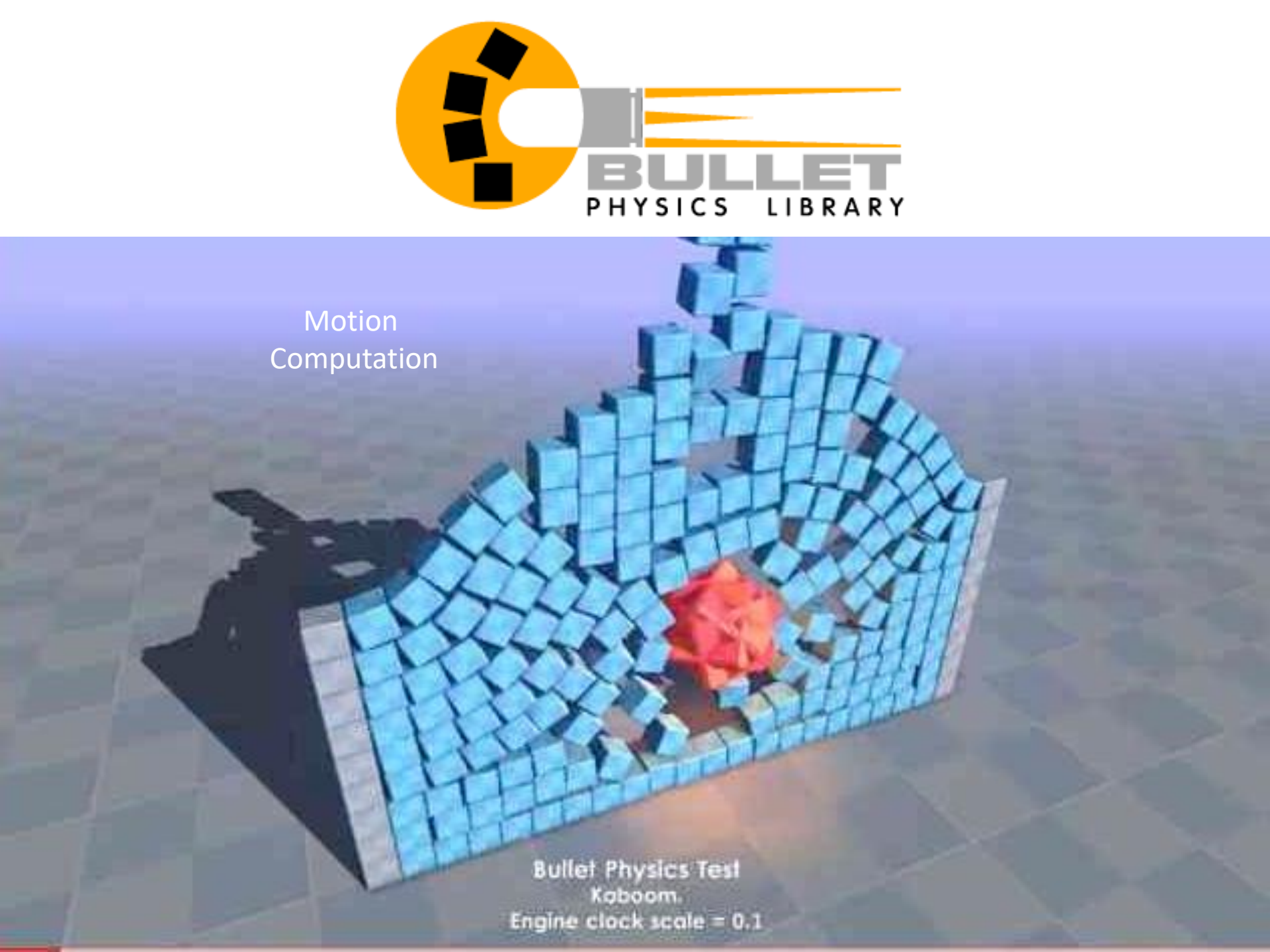
	Initial Reconstruciton using Camera 1	Refined Reconstruction using Camera 1	Initial Reconstruction using Camera 1 & 2	Initial Reconstruciton using Camera 1	Refined Reconstruction using Camera 1	Initial Reconstruction using Camera 1 & 2
View from Camera 1						
View from Camera 2						
Vlew from Camera 3						

Deformation

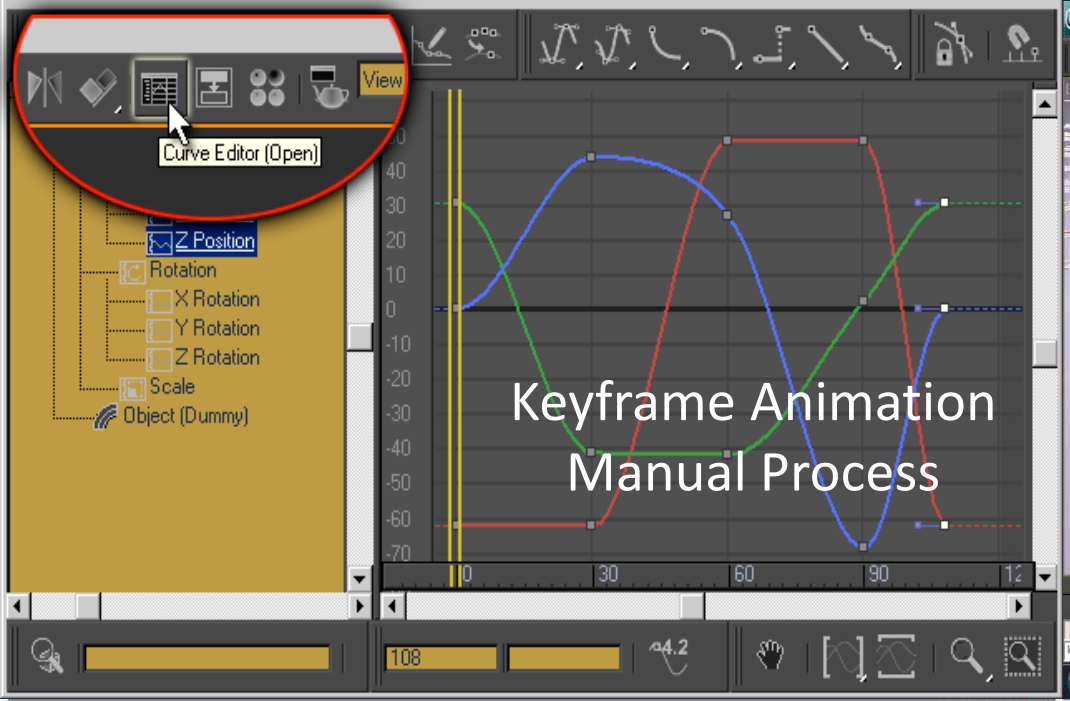


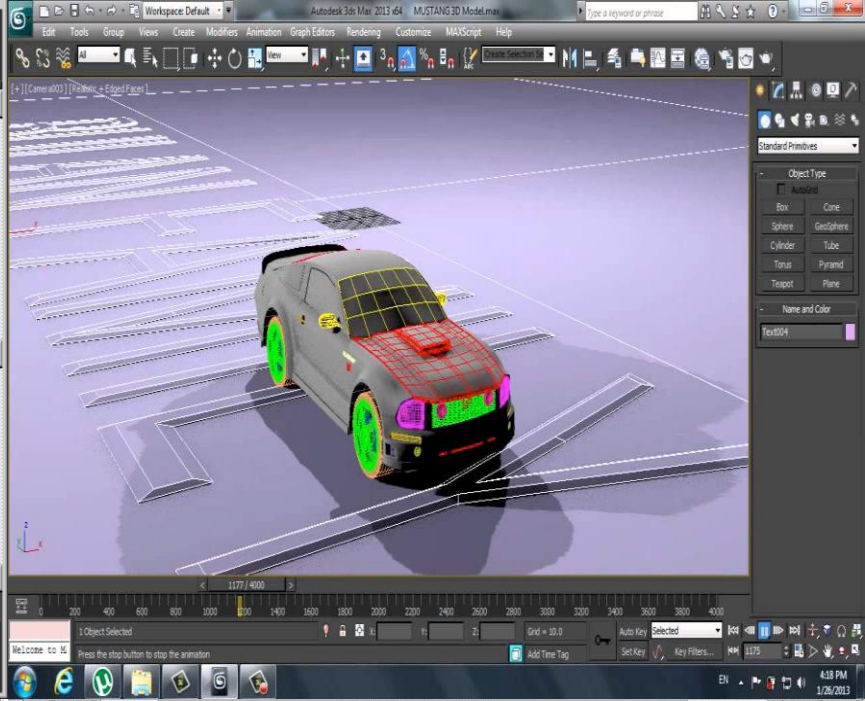

[Paused]
Resume Demo
Restart Demo
Toggle Text Statistics
Toggle Graph Statistics
Toggle Camera Info
Settings
Toggle Help
Single Step
Quit
Motion
Computation

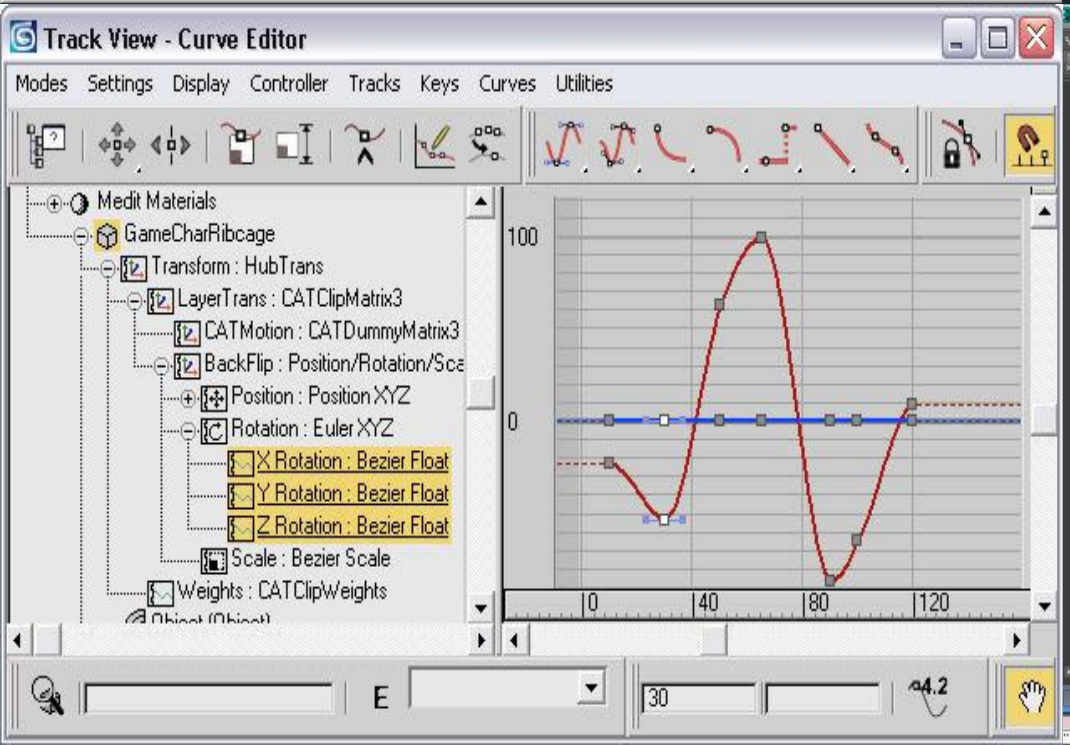

geekswipe.net
Motion
Computation
PhysX
by NVIDIA


BULLET
PHYSICS LIBRARY
Motion
Computation
Bullet Physics Test
Kaboom.
Engine clock scale = 0.1

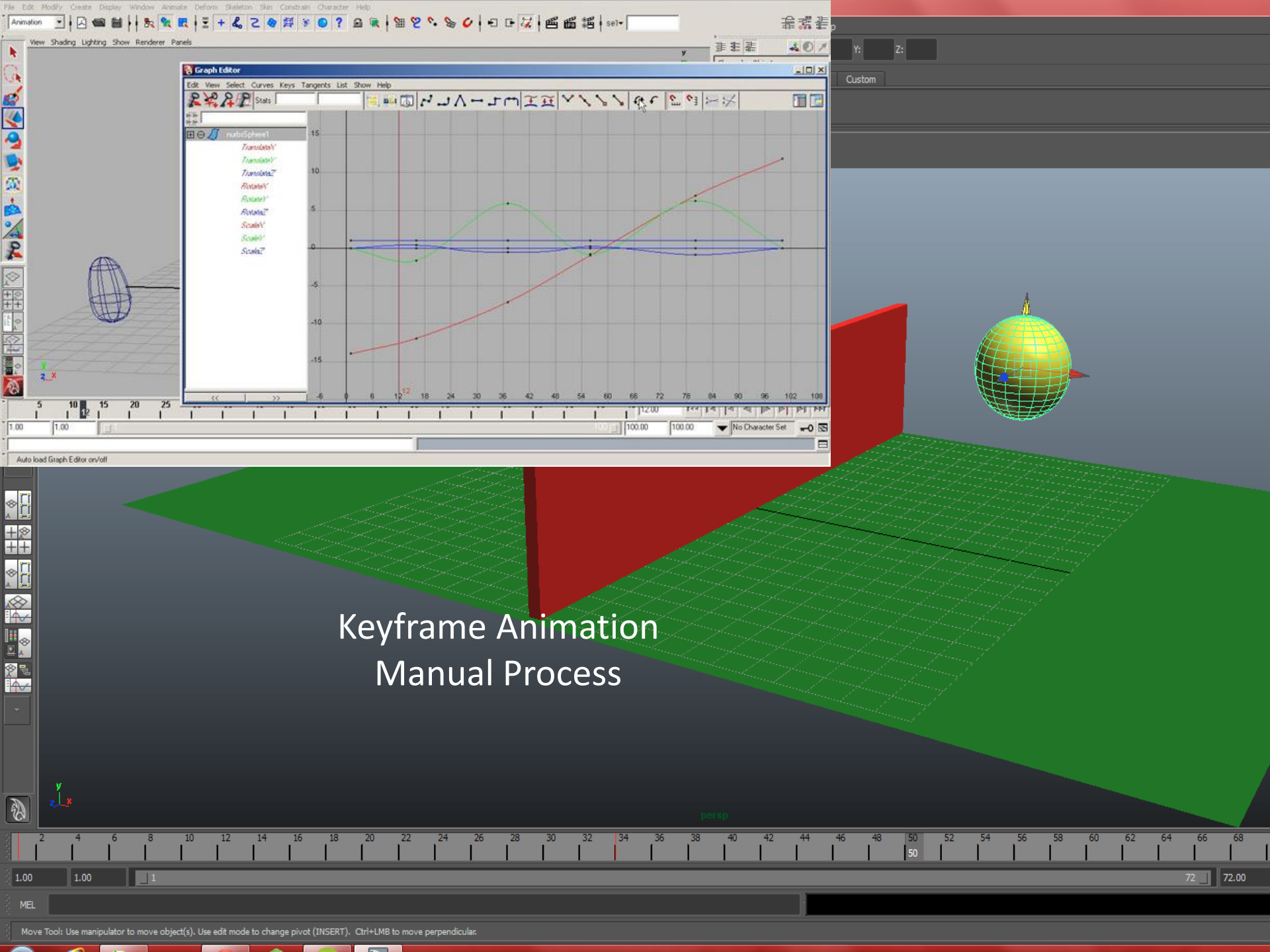

havok™
Graphite Modeling Tools
Freeform
Selection
Object Paint
Polygon Modeling
[+] [Perspective] [Realistic]
Modifier List
Motion
Computation
26 / 250
None Selected
Click or click-and-drag to select objects
Add Time Tag
Auto Key
Set Key
Key Filters...


View
Curve Editor (Open)
Z Position
Rotation
X Rotation
Y Rotation
Z Rotation
Scale
Object (Dummy)
Keyframe Animation
Manual Process

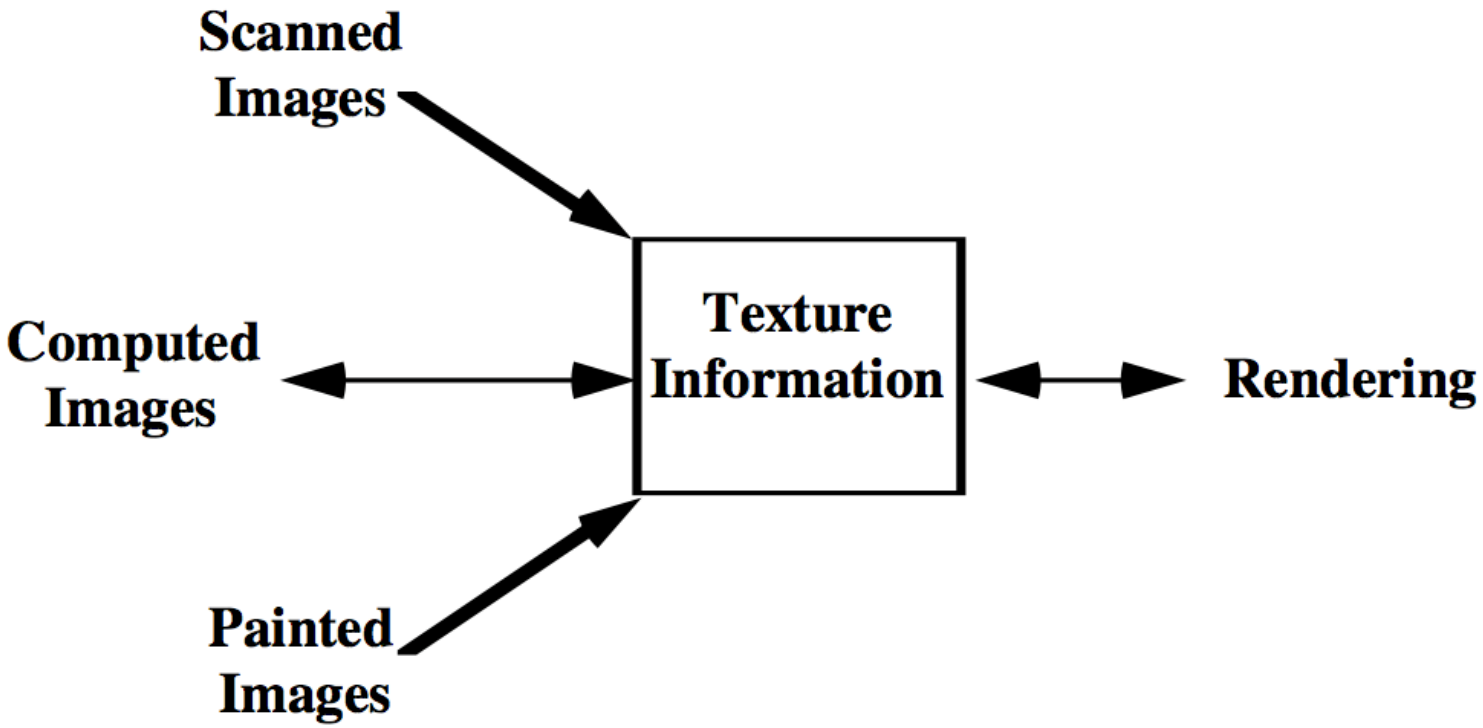



Track View - Curve Editor
Modes Settings Display Controller Tracks Keys Curves Utilities
Medit Materials
GameCharRibcage
Transform : HubTrans
LayerTrans : CATClipMatrix3
CATMotion : CATDummyMatrix3
BackFlip : Position/Rotation/Sca
Position : Position XYZ
Rotation : Euler XYZ
X Rotation : Bezier Float
Y Rotation : Bezier Float
Z Rotation : Bezier Float
Scale : Bezier Scale
Weights : CATClipWeights

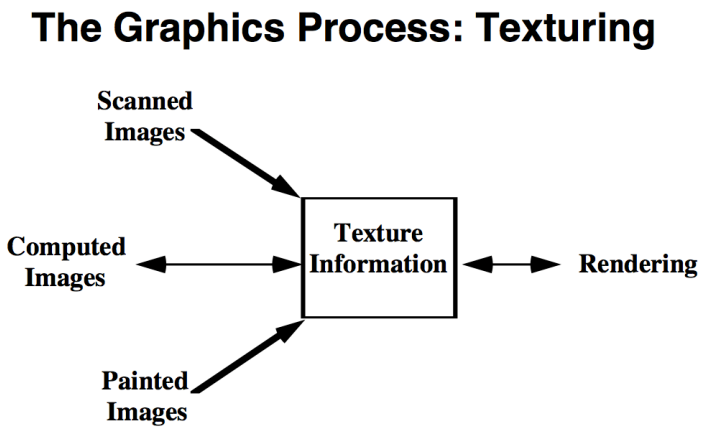



Graph Editor
Edit View Select Curves Keys Tangents List Show Help
Custom
Keyframe Animation
Manual Process
MEL
Move Tool: Use manipulator to move object(s). Use edit mode to change pivot (INSERT). Ctrl+LMB to move perpendicular.

The Graphics Process: Texturing

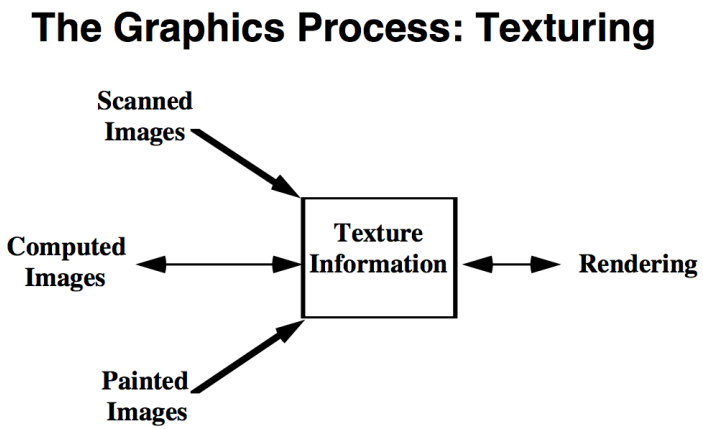


The Graphics Process: Texturing



- **Texture Information** is the process of applying realistic surface details to 3D models during **rendering**.

The Graphics Process: Texturing

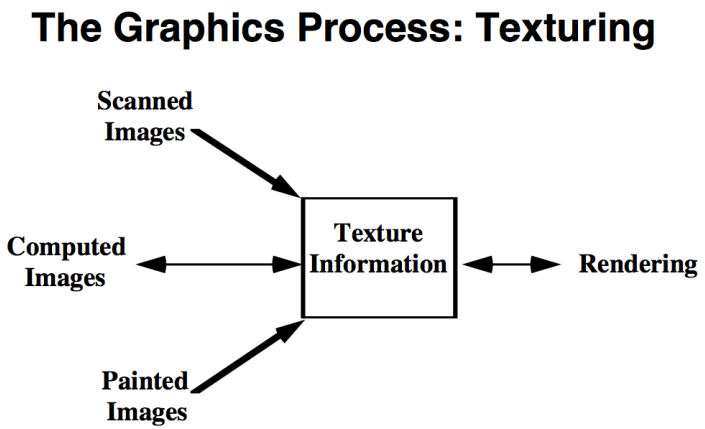


Inputs to Texture Information:

Scanned Images:

1. These are real-world images captured via scanners or cameras.
2. They are used as textures to give realism to objects (e.g., a photo of a brick wall applied to a 3D wall).

The Graphics Process: Texturing

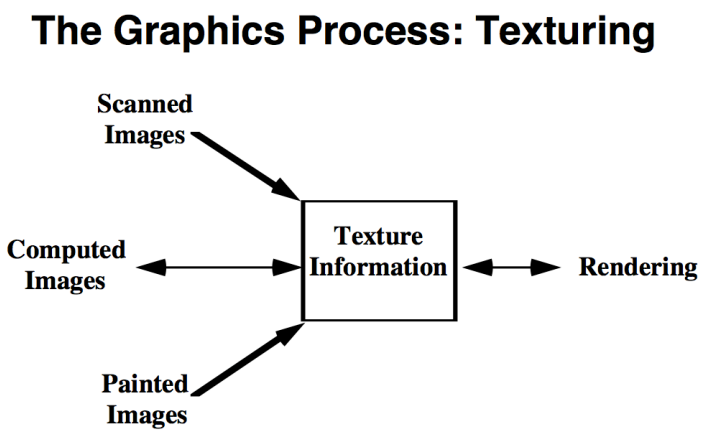


Inputs to Texture Information:

Computed Images:

1. These are textures generated algorithmically or through simulations (e.g., procedural textures like marble, clouds, or noise).
2. Useful for patterns that are hard to draw manually or photograph.

The Graphics Process: Texturing

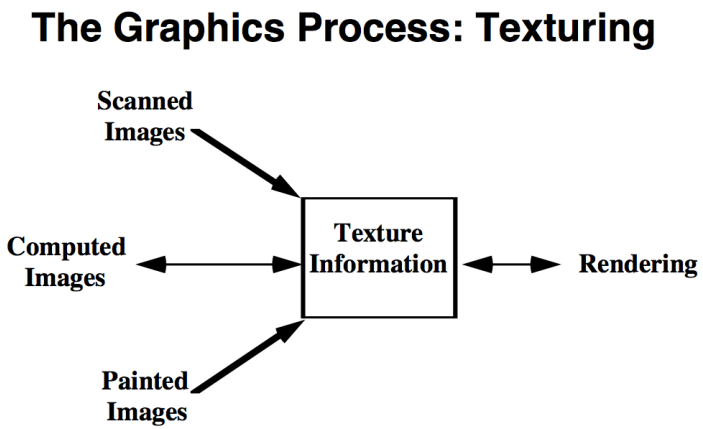


Inputs to Texture Information:

Painted Images:

1. These are manually created textures using digital painting tools (e.g., Photoshop, Substance Painter).
2. Artists can paint directly on 2D textures or even onto 3D models.

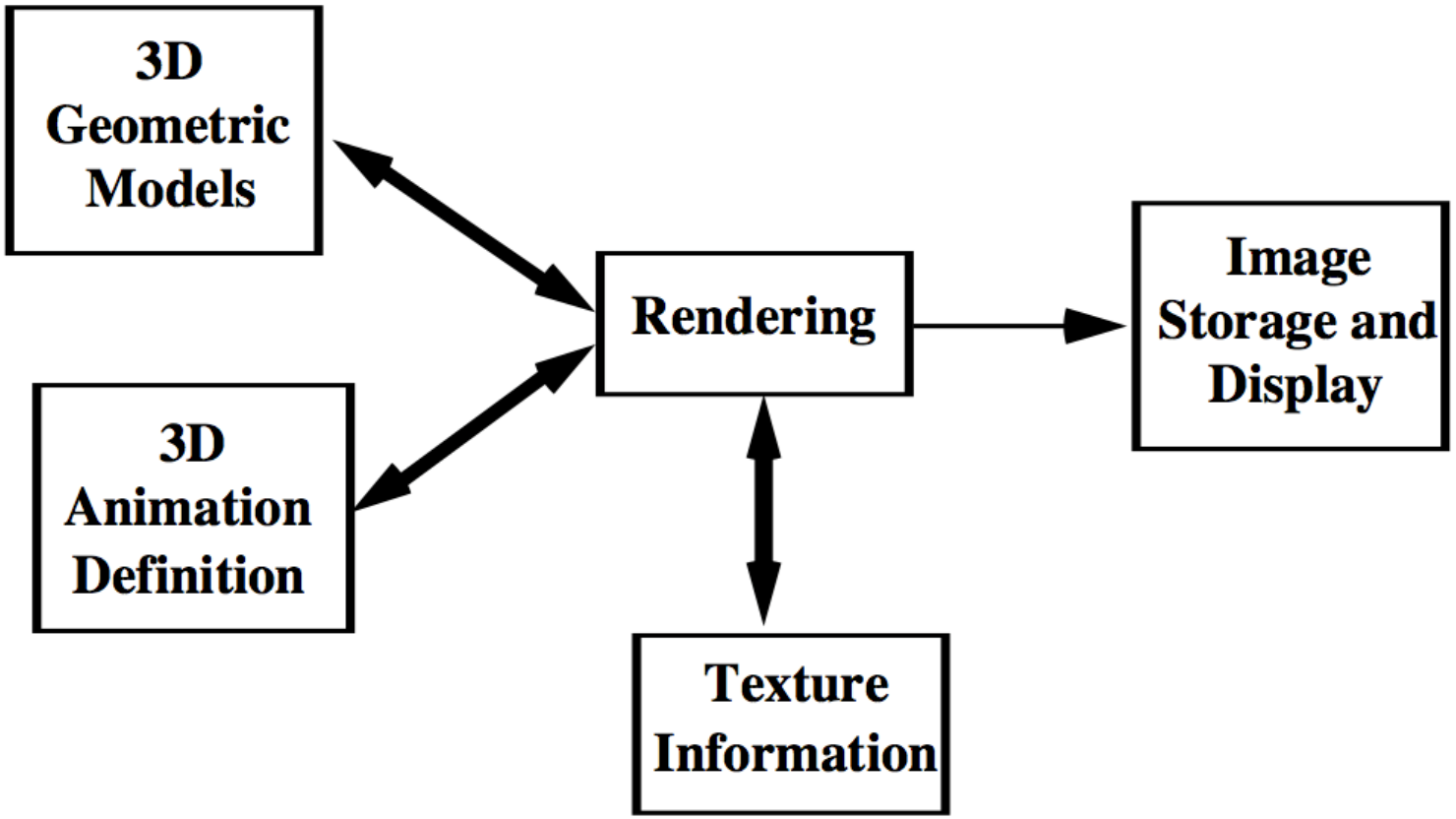
The Graphics Process: Texturing



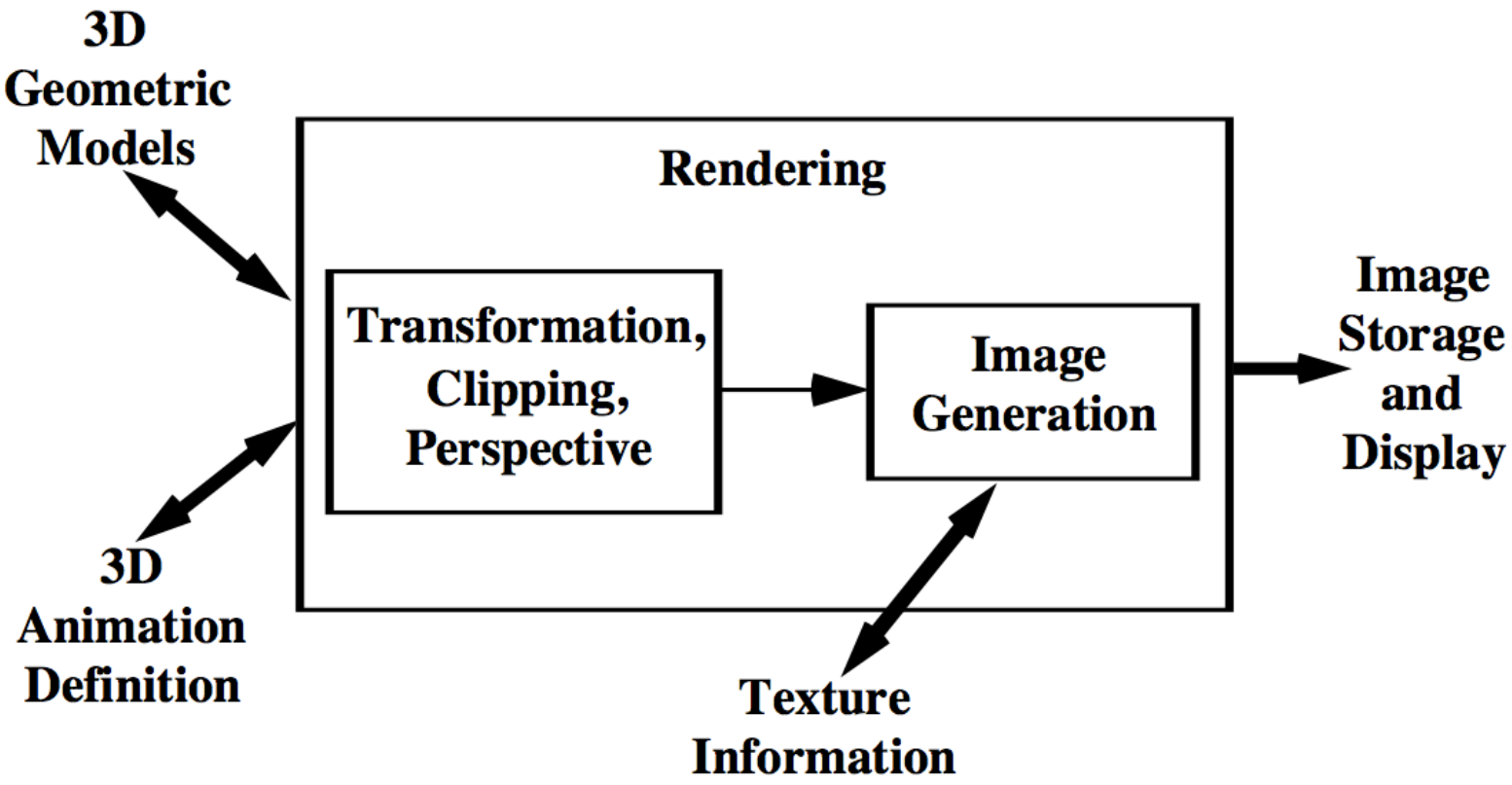
Output from Texture Information:

- **Rendering**:
  - During rendering, the texture information is mapped onto 3D models.
  - It adds detail, realism, and visual richness to objects in the final image.

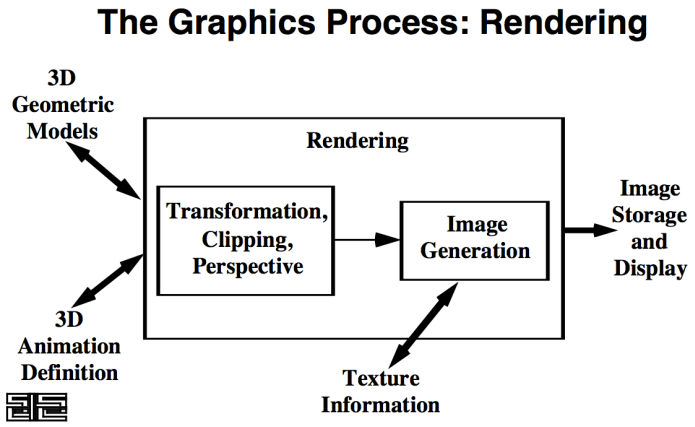
The Graphics Process: Summary



The Graphics Process: Rendering



The Graphics Process: Rendering



Transformation, Clipping, Perspective

- This step includes key stages of 3D graphics rendering:
- **Transformation**: Converts 3D model coordinates into world space, camera space, and finally screen space.
- **Clipping**: Removes objects or parts of objects that are outside the camera's view.
- **Perspective**: Applies the camera's lens effect to simulate depth (making distant objects appear smaller).

The Graphics Process: Rendering

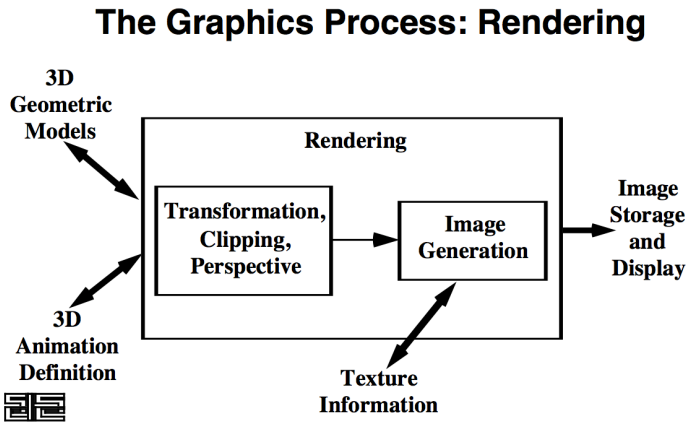


Image Generation

- This is where the actual **pixels** of the image are calculated. It involves:
- Applying lighting models
- Shading surfaces
- Mapping textures (using texture info)
- Performing visibility checks (like z-buffering)

The Graphics Process: Rendering

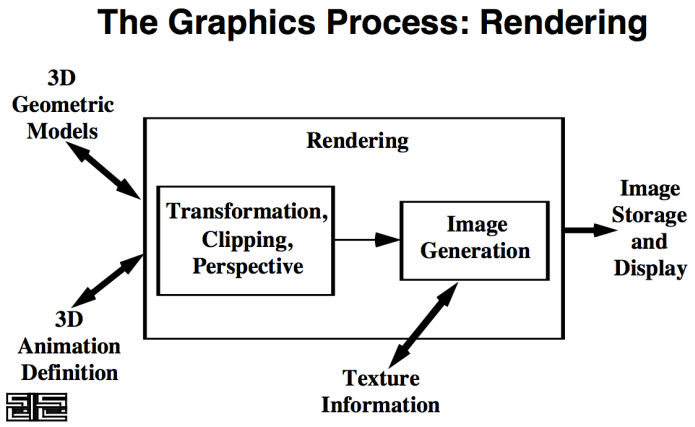


Image Storage and Display

- Once the image is generated, it is stored in a frame buffer and then displayed on a screen or saved as a file.

Z-buffering

- **Z-buffering:** (also known as depth buffering) is a computer graphics technique used to determine which objects, or parts of objects, are visible in a 3D scene and which are hidden behind others from the viewer's perspective.

- **Z-fighting:** Occurs when two surfaces are extremely close together, and the Z-buffer can't decide which one is in front due to limited precision.

Z-buffering How: Z-Buffering Works

- Each pixel on the screen has a corresponding Z-buffer value (depth value), which stores the distance from the camera to the closest object at that pixel.
  - When rendering:
    - For every fragment (potential pixel) generated, the graphics system calculates its depth (Z value).
    - The fragment's Z value is compared with the current Z-buffer value at that pixel.
  - Comparison:
    - If the new Z value is closer (smaller), the pixel is updated with the new color and the Z-buffer is updated with this new depth.
    - If it's distant, the pixel is discarded (something is already in front of it).